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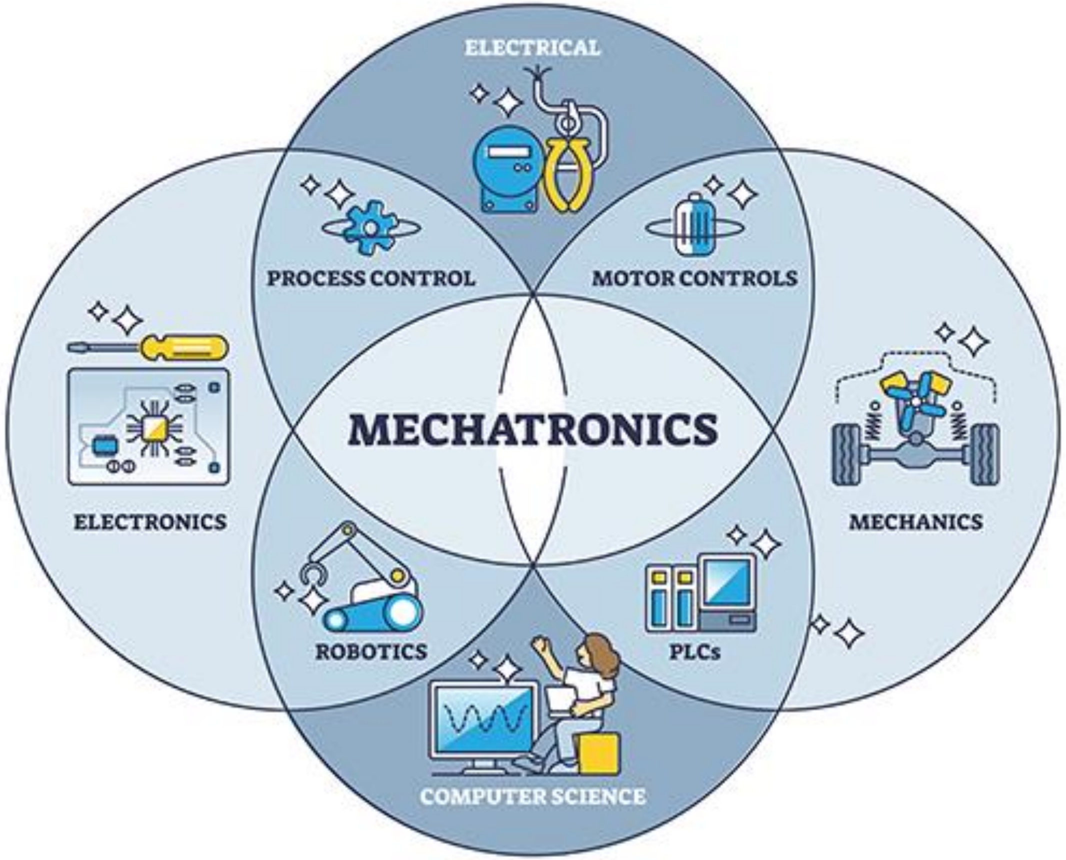
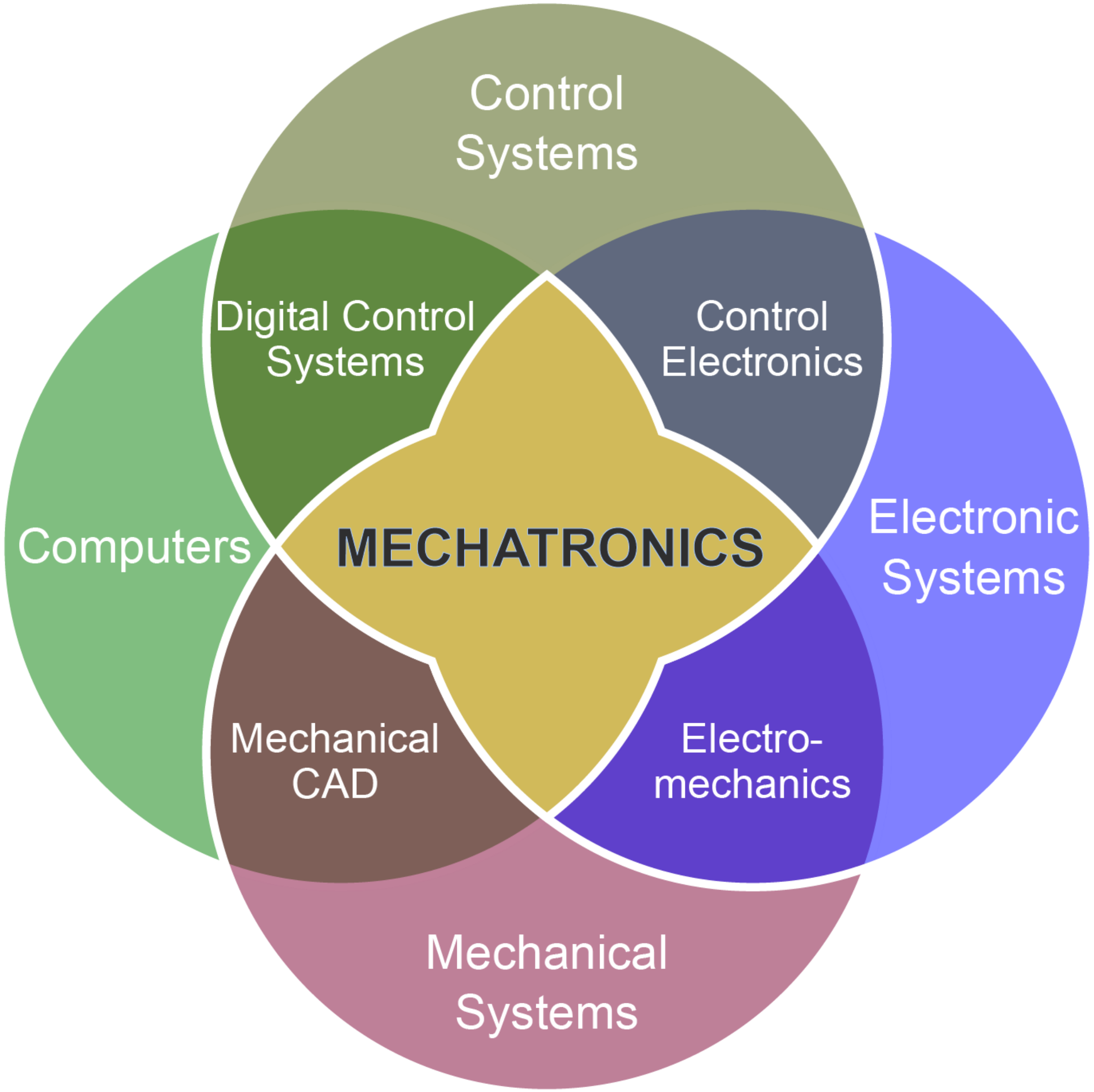
Mechatronics

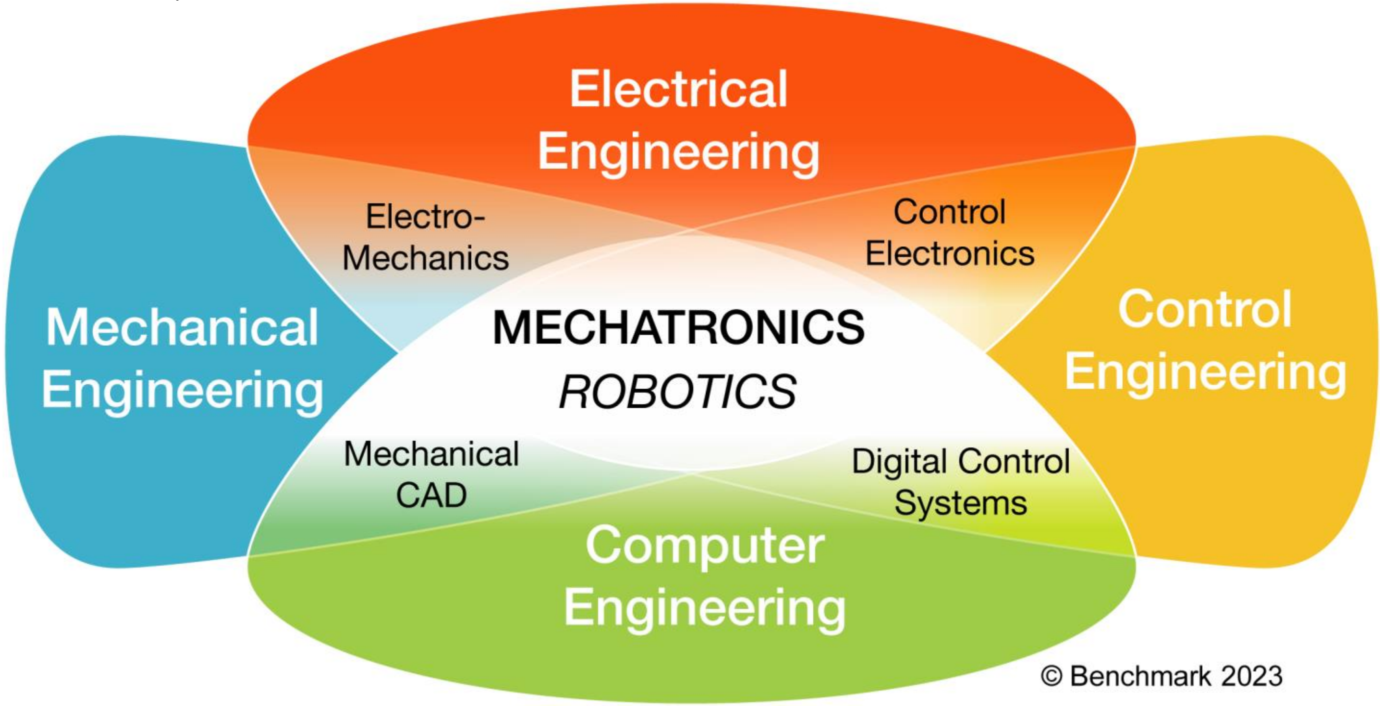


Introduction



Introduction

- Mechatronics is an interdisciplinary field of engineering that combines mechanical engineering, electrical engineering, computer science, and control engineering to design and build intelligent products.
- In simpler terms, it's about merging the various engineering disciplines to create smart systems that can sense, think, and act

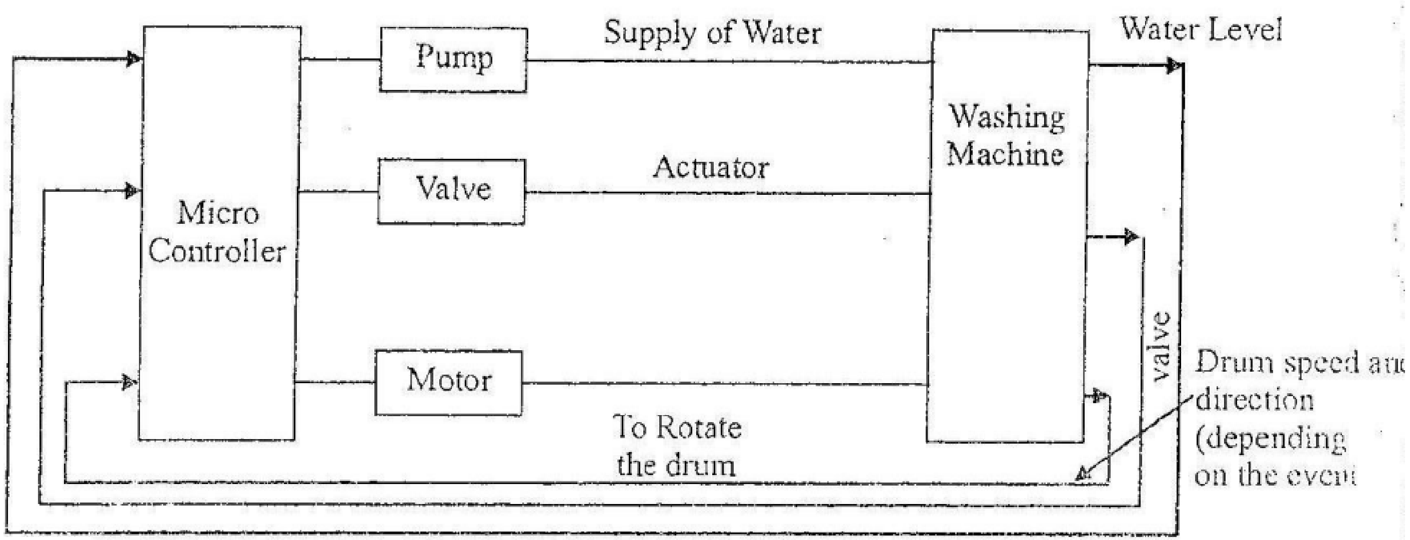


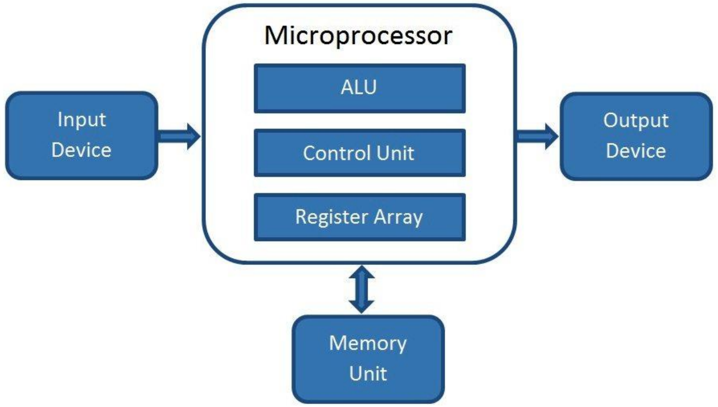
To define mechatronics with single sentence

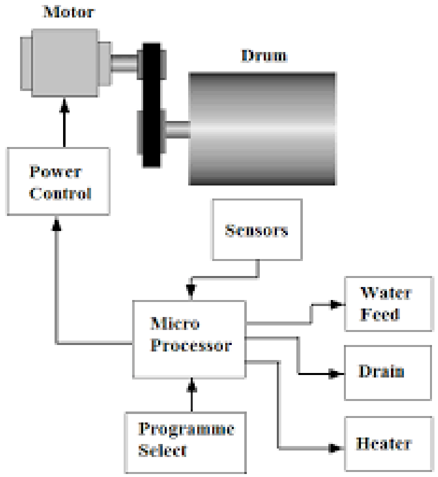
- *Mechatronics is the synergistic integration of mechanical engineering with electronics and electrical with intelligent computer control in the design and manufacture of industrial products, processes and operations*

Class room activity

Block diagram of a microprocessor based processor control system of Automatic washing machine







Scope of Mechatronics

Mechatronics plays a vital role in industrial sector

- Better design of products. In mechatronics, computer is the key constituent. Computer- aided designing (CAD) involves the use of computers to assist in designing of an individual part, machine tool, product or system in three stages-conceptual, preliminary and final.
- Better process planning. It deals with the use of computers in process planning, i.e. computer-aided process planning (CAPP). It helps lower manufacturing costs and higher product quality.
- Reliable and quality-oriented manufacturing. Because of computer-integrated manufacturing (CIM), the reliable and high quality-oriented products can be manufactured.
- Intelligent process control. Due to developments in digital computer systems, their use in process control has extensively increased. In power plants, process and manufacturing industries, computer-aided process control is used for passive and active applications.
- Passive applications include monitoring and alarming of data from various processes.
- Active applications involve acquisition and manipulation with additional features like process control, process and plant optimization and tuning of various controllers for best operating performance. Nowadays, in mechatronic systems, artificial neural networks are used in manufacturing systems for process control and inspection to improve production performance and production quality.

In general, mechatronics helps industries achieve greater productivity, reliability and higher quality by incorporating intelligent self-correcting sensory and feedback system.

Conventional Design and Mechatronics Design

Convectional design	Mechatronics Design
Bulky componentized systems	Compact integrated systems
Complex mechanism; Complex mechanical mechanisms	Simple Mechanism: Replacement of many complex mechanical components and/or systems with electronic, computer and/or software systems
Cable problem	Bus or wireless communication
Simple control • Stiff construction • Feedback control, linear(analog) control • Precision through narrow tolerance • Non measurable quantities changes arbitrarily • Simple monitoring • Fixed abilities	Integration by information processing(software) • elastic construction with damping by electronic feedback • programmable feedback, (nonlinear) digital control • precision through measurement and feedback control • control of non-measurable estimated quantities • supervision over fault diagnosis • learning ability
Centralized processing & control	Hybrid Control: Adaptive and/or Multi-architecture control (e.g., Centralized, Centralized processing & control Decentralized and Distributed)
Constant speed drives	Variable speed drives
Mechanical Systems	Mechanical, Computer, Electronic, Software, and/or Network interface and/or control of physical, chemical, biological and/or neurological systems

System

- A system can be thought of as a box which has an input and an output and where we are not concerned with what goes on inside the box but only the relationship between the output and the input.

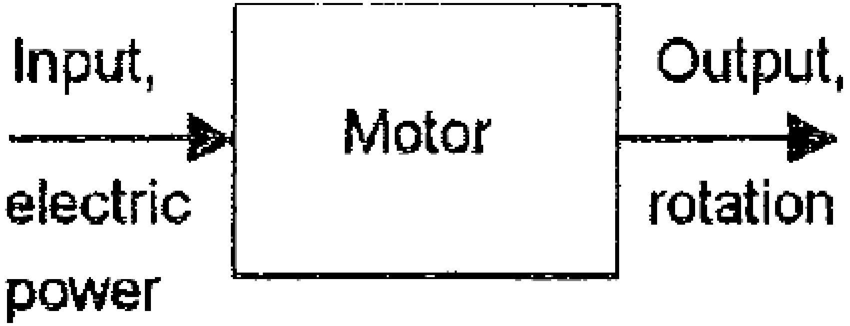


Fig. 1.1 An example of a system

- input electric power and as output the rotation of a shaft

Measurement system

- A measurement system can be thought of as a black box which is used for making measurements. It has as its input the quantity being measured and its output the value of that quantity

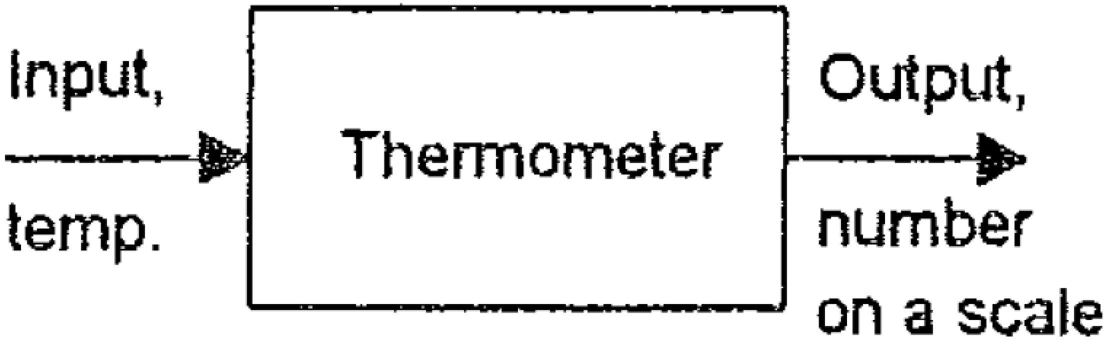


Fig. 1.2 An example of a measurement system

control system

- A control system can be thought of as a black box which is used to control its output to some particular value or particular sequence of values.

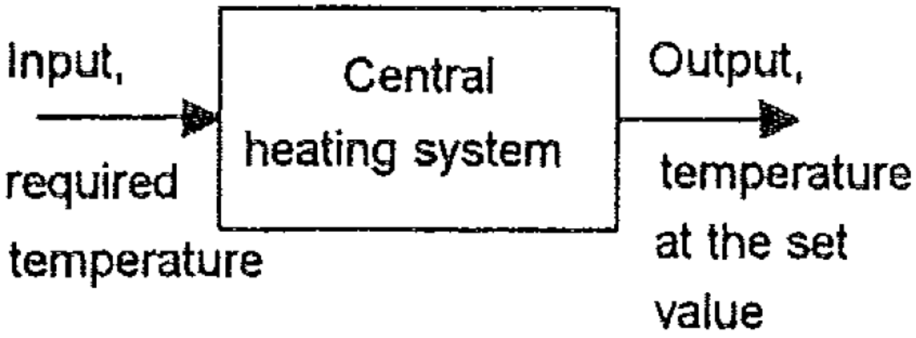
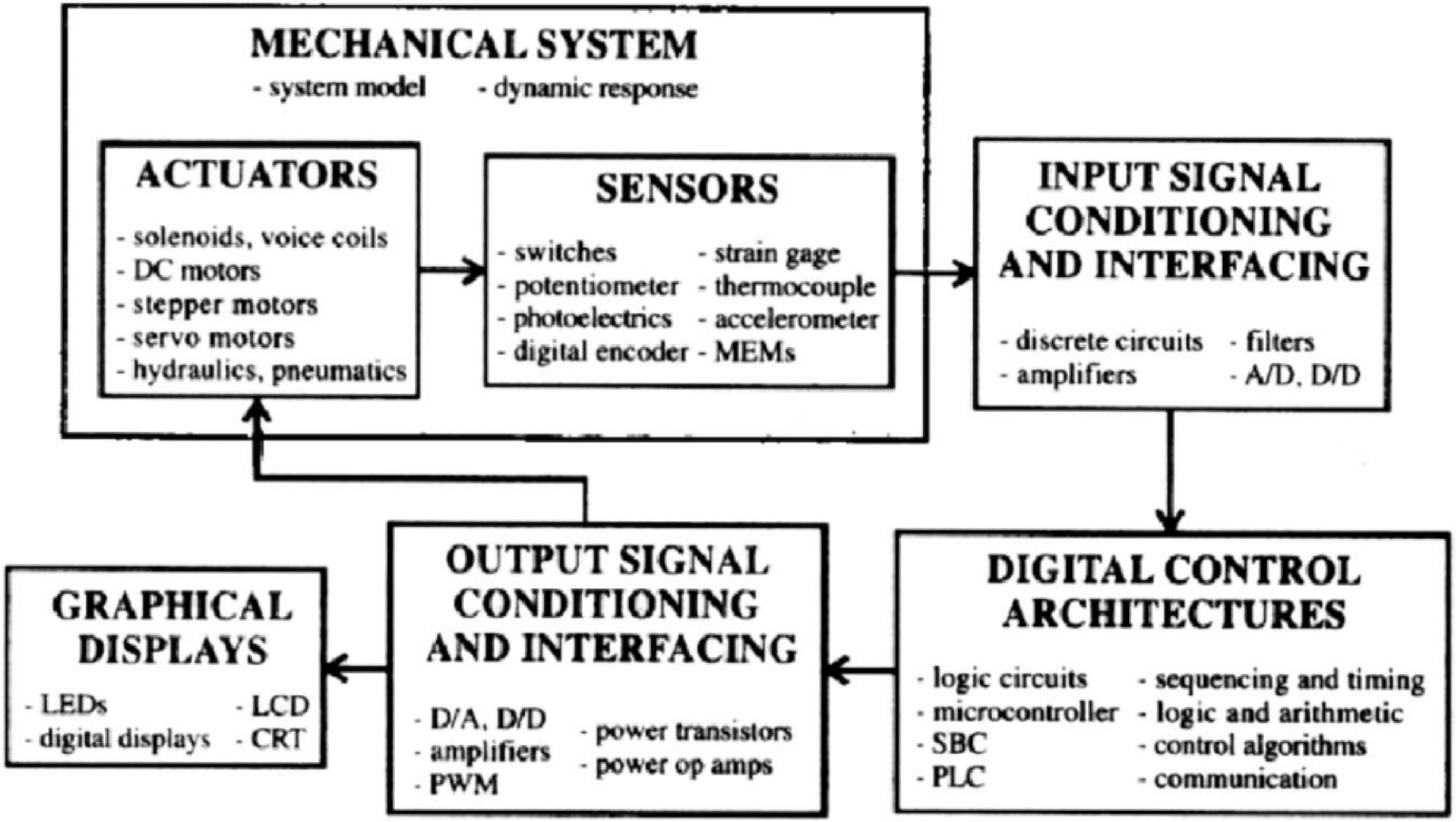


Fig. 1.3 An example of a control system

- a domestic central heating control system has as its input the temperature required in the house and as its output the house at that temperature

Key elements of Mechatronics system



Sensor and transducer

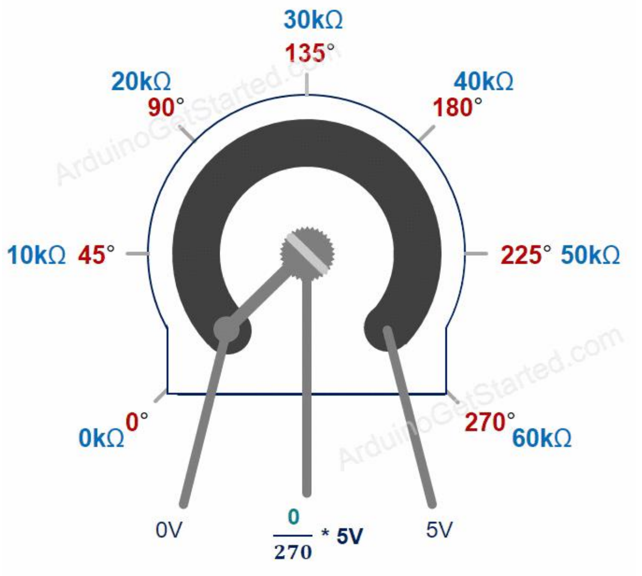
- The heart of any measurement or control system is sensor/transducer.
- Sensor/transducer is a device it converts the one form of energy to another form.
- Sensor/transducer it senses the physical phenomenon to be measure and transform it from one form to another form (generally electrical form)
- *Sensor is a device when exposed to a physical phenomenon (temperature, displacement, force, etc.) produces a proportional output signal (electrical, mechanical, magnetic, etc.)*
- a sensor is a device that responds to a change in the physical phenomenon
- a transducer is a device that converts one form of energy into another form of energy
- Ex: Thermocouple responds to a temperature change (thermal energy) and outputs a proportional change in electromotive force (electrical energy). Therefore, *a thermocouple can be called a sensor and or transducer*

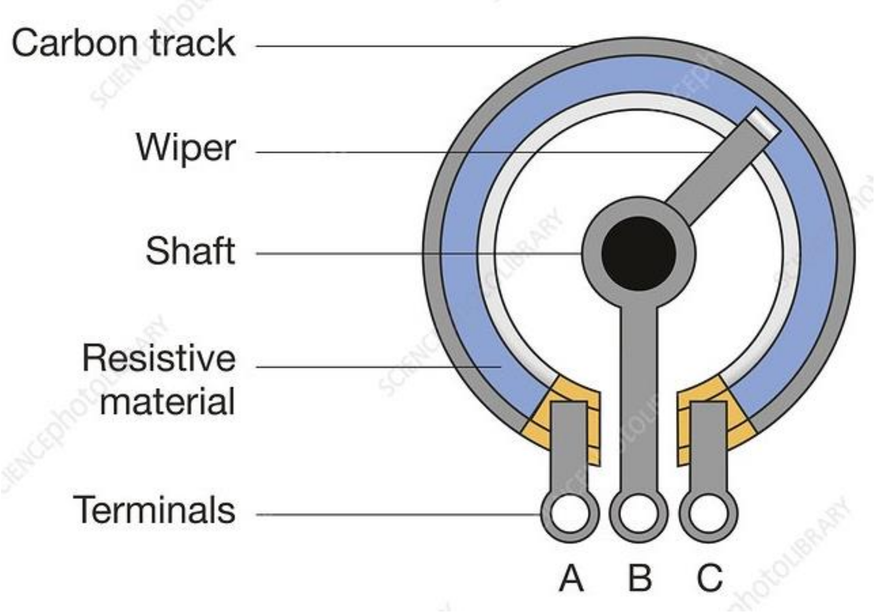
1. Displacement, position and proximity

- Displacement sensors - the measurement of the amount by which some object has been moved
- Position sensors - the determination of the position of some object in relation to some reference point
- Proximity sensors are a form of position sensor - a device that can detect the presence of nearby objects without any physical contact.

1.1 Potentiometer sensor

- A potentiometer consists of a resistance element with a sliding contact which can be moved over the length of the element.
- Such elements can be used for linear or rotary displacements, the displacement being converted into a potential difference.
- It works on the principle of change of resistance of the wire with its length
- Convert rotary or linear displacement to a voltage to measure angular and linear position.
- Through voltage division, *the change in resistance can be used to create an output voltage that is directly proportional to the input displacement*





Potentiometer sensor

- The rotary potentiometer consists of a circular wire-wound track or a film of conductive plastic over which a rotatable sliding contact can be rotated
- With a constant input voltage V_s between terminals 1 and 3, the output voltage V_o between terminals 2 and 3 is a *fraction of the input voltage*
- The fraction depending on the ratio of the resistance R_{23} between terminals 2 and 3 compared with the total resistance R_{13} between terminals 1 and 3

$$\frac{V_0}{V_s} = \frac{R_{23}}{R_{13}}$$

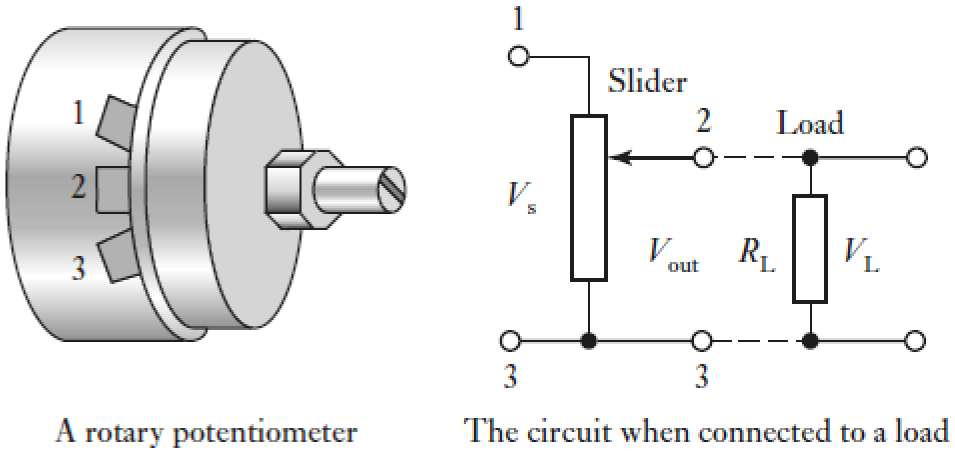


Figure 2.5 Rotary potentiometer.

Effect to be considered with a potentiometer

- a load R_L connected across the output
- The potential difference across the load V_L
- The potentiometer resistance $R_P = R_L$ * (fraction **x)**
- the error introduced by the load having a finite resistance is ***(error in potentiometer due to the deviation of resistance output with wiper position)***

$$error = V_s \frac{R_P}{R_L}(x^2 - x^3)$$

To illustrate the above, consider the non-linearity error with a potentiometer of resistance 500 Ω, when at a displacement of half its maximum slider travel, which results from there being a load of resistance 10 kΩ. The supply voltage is 4 V. Using the equation derived above

$$\text{error} = 4 \times \frac{500}{10\,000}(0.5^2 - 0.5^3) = 0.025 \text{ V}$$

Capacitive element

The capacitance C of a parallel plate capacitor is given by

$$C = \frac{\varepsilon_r \varepsilon_0 A}{d}$$

- ε_r = relative permittivity of the dielectric
- ε_0 = permittivity of free space
- A= area of overlap between the two plates
- d = plate separation

By changing ε_r, A, d we can change the capacitance and hence cab be used in sensors for sensing purpose

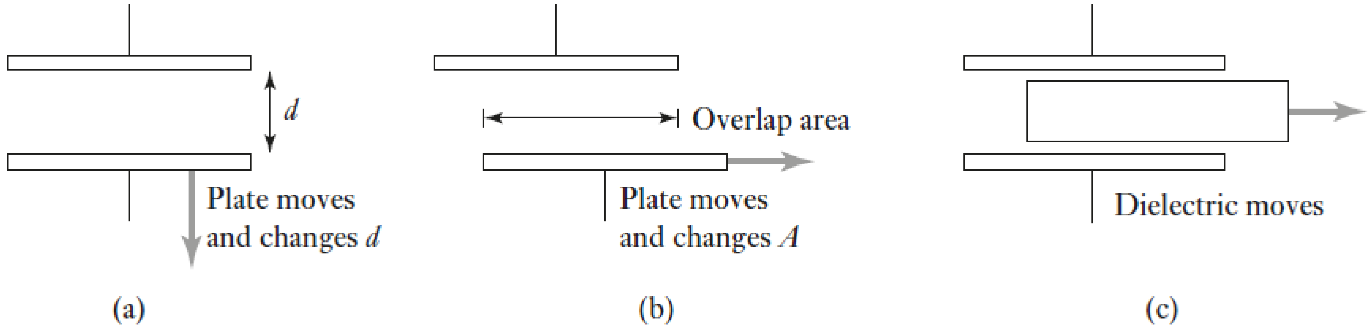
Capacitive sensors for the monitoring of linear displacements in different forms

a) One of the plates is moved by the displacement so that the plate separation changes

b) The displacement causes the area of overlap to change

c) The displacement causes the dielectric between the plates to change

Figure 2.8 Forms of capacitive sensing element.



Strain-gauged element

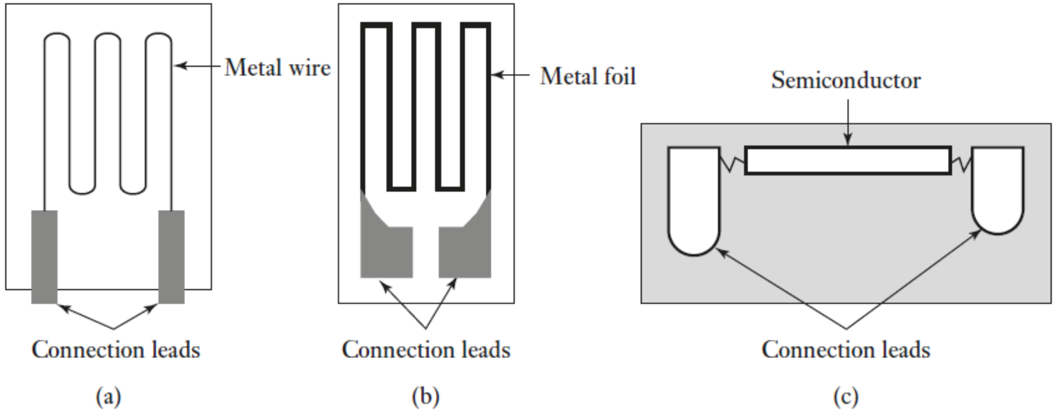
- A strain-gauged element is a sensor that converts mechanical strain (deformation) into an electrical signal
- The electrical resistance strain gauge is a metal wire, metal foil strip or semiconductor strain gauge

When the object experiences mechanical strain (stretching, compressing, bending, etc.), the strain gauge deforms along with it. Due to the deformation, the length and cross-sectional area of the metallic foil change, which in turn causes its electrical resistance to change. This change in resistance is proportional to the applied strain.

- When subject to strain, its resistance R changes, the fractional change in resistance $\frac{\Delta R}{R}$ being proportional to the strain $\boldsymbol{\varepsilon}$

$$\frac{\Delta R}{R} = G\varepsilon$$

Figure 2.6 Strain gauges: (a) metal wire, (b) metal foil, (c) semiconductor.



where G, the constant of proportionality, is termed the gauge factor

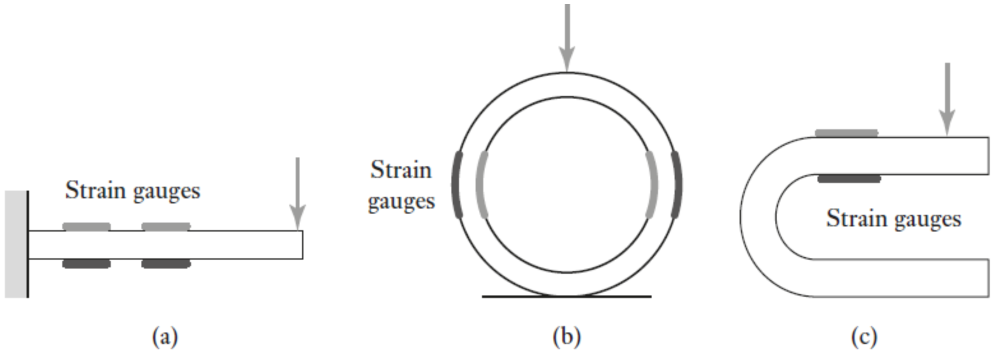
Strain-gauged element

Since strain is the ratio (change in length/original length) then the resistance change of a strain gauge is a measurement of the change in length of the element to which the strain gauge is attached.

- A problem with all strain gauges is that their resistance changes not only with strain but also with temperature.
- Semiconductor strain gauges have a much greater sensitivity to temperature than metal strain gauges
- strain gauges attached to flexible elements in the form of cantilevers, rings or U-shapes. When the flexible element is bent or deformed as a result of forces being applied by a contact point being displaced, then the electrical resistance strain gauges mounted on the element are strained and so give a resistance change which can be monitored.

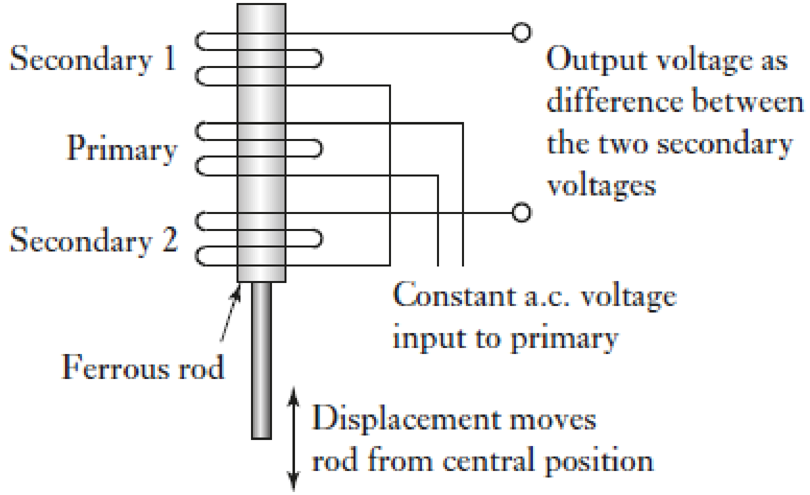
To illustrate the above, consider an electrical resistance strain gauge with a resistance of 100 Ω and a gauge factor of 2.0. What is the change in resistance of the gauge when it is subject to a strain of 0.001?

$$\text{change in resistance} = 2.0 \times 0.001 \times 100 = 0.2\ \Omega$$

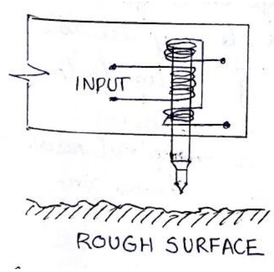


Differential transformers : Linear Variable Differential Transformer (LVDT)

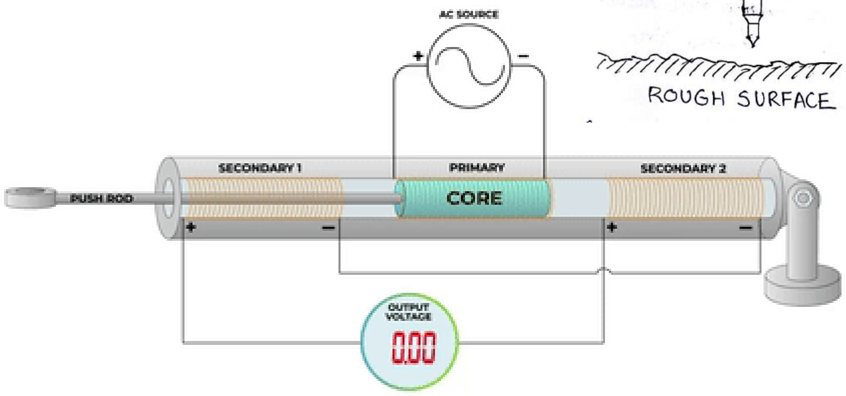
- It is inductive sensor
- Magnitude of the output voltage $\propto$ The displacement of the ferrite core within the primary and secondary winding
- LVDT consists of three coils symmetrically spaced along an insulated tube



- The central coil is the primary coil and the other two are identical secondary coils which are connected in series
- *When there is an alternating voltage input to the primary coil, alternating emf induced in the secondary coils which is equal in magnitude and opposite in direction.*



- When the core within it changes the position, the voltage induced gets changed (e.g. more in secondary coil 2 than coil 1. The result is that a greater emf is induced in one coil than the other)

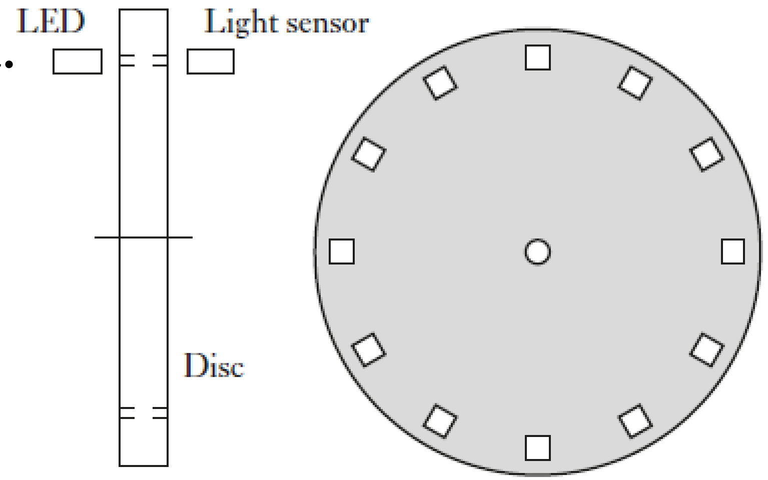


- This principle is used in measuring the roughness of the surface

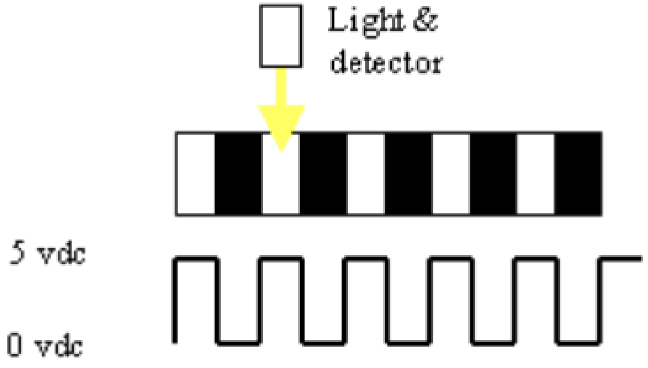
Optical encoders

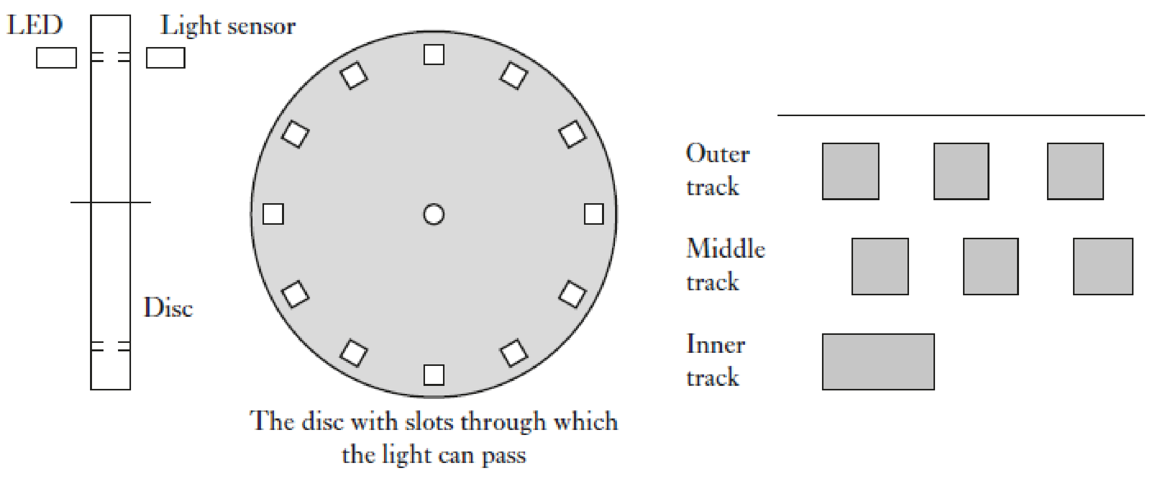
- *An encoder is a device that provides a digital output as a result of a linear or angular displacement*
- Encoders can be grouped into two categories: *incremental encoders and absolute encoders*
- Incremental encoders: detect ***changes in position*** (rotation) from some datum by generating pulses with each movement
- Absolute encoders : give the ***actual angular position*** at any given moment
- Incremental encoder for the measurement of angular displacement.
- A beam of light passes through slots in a disc and is detected by a suitable light sensor.
- When the disc is rotated, a pulsed output is produced by the sensor.

The number of pulses proportional to the angle through which the disc rotates

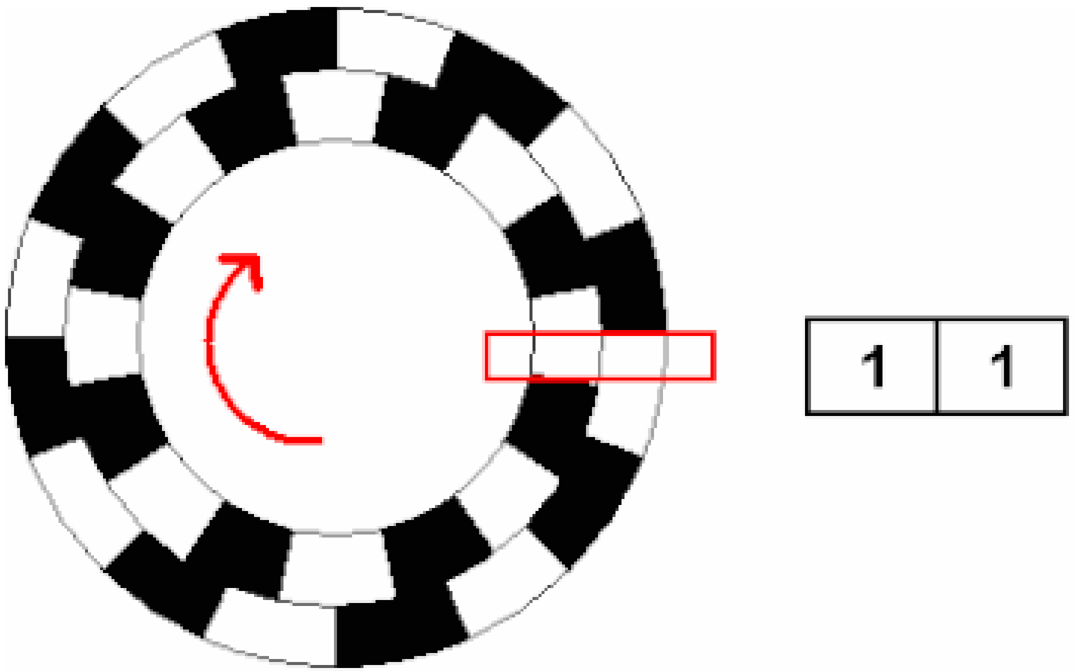


The disc with slots through which the light can pass





(a) (b)



- The outer and middle tracks have a series of equally spaced holes that go completely round the disc but with the holes in the middle track offset from the holes in the outer track by one-half the width of a hole.
- This offset enables the direction of rotation to be determined.
- In a ***clockwise direction the pulses in the outer track lead those in the inner***
- In the anti-clockwise direction they lag

Pneumatic sensors

- Pneumatic sensors involve the use of compressed air
- *The displacement or the proximity of an object being transformed into a change in air pressure*
- Low pressure air is allowed to escape through a port in the front of the sensor, this escaping air, in the absence of any close-by object
- *If there is a close-by object, the air cannot so readily escape and the result is that the pressure increases in the sensor output port*
- This increase in pressure depends on the proximity of objects

sensors are used for the measurement of displacements of fractions of millimetres in ranges which typically are about 3 to 12 mm.

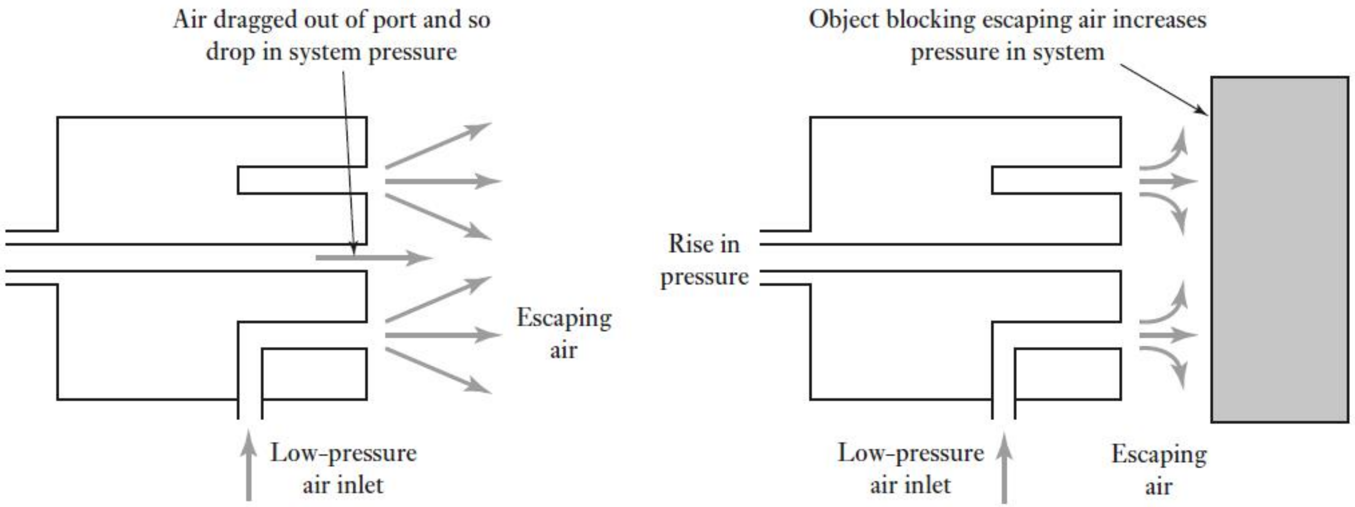
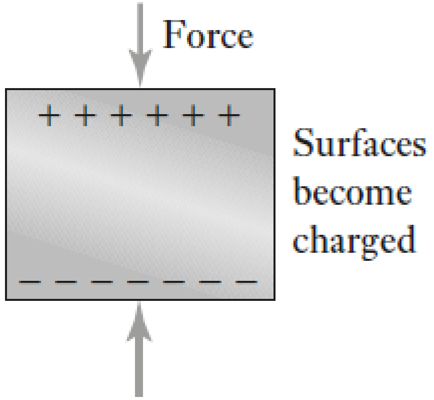


Figure 2.19 Pneumatic proximity sensor.

Piezoelectric sensors

- Piezoelectric materials when *stretched or compressed* generate electric charges
- one face of the material becoming positively charged and the opposite face negatively charged, as a result, a voltage is produced



- Piezoelectric materials are ionic crystals, which *when stretched or compressed* result in the *charge distribution* in the crystal changing so that there is a *net displacement of charge* with one face of the material becoming positively charged and the other negatively charged.

Piezoelectric sensors

- The net charge q on a surface is proportional to the amount *x* by which the charges have been displaced, and since the displacement is proportional to the applied force *F*

$$q = kx = SF$$

where *k* is a constant and *S* a constant termed the charge sensitivity

- Quartz has a charge sensitivity of 2.2 pC/N
- Barium titanate has a much higher charge sensitivity of 130 pC/N
- lead zirconate–titanate about 265 pC/N

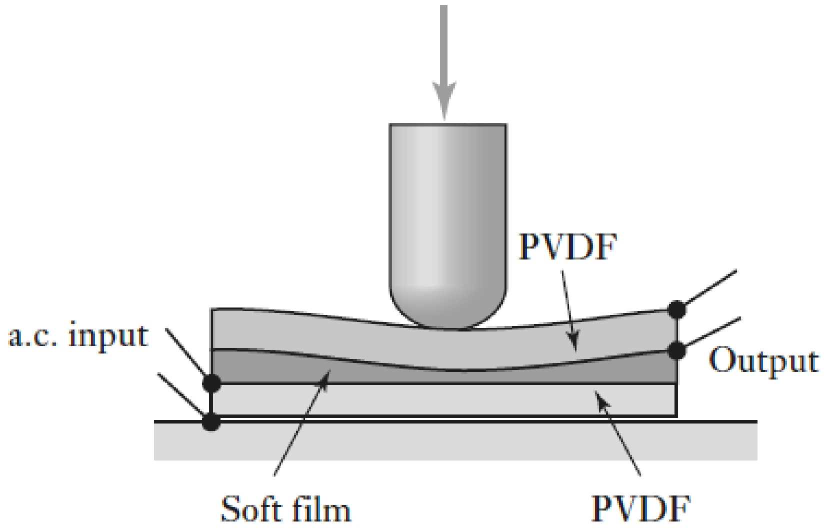
Tactile sensor

- A tactile sensor is a form of pressure sensor.
- Tactile sensor is used on the ‘fingertips’ of robotic hands to determine when a hand has come into contact with an object.

Piezoelectric polyvinylidene fluoride (PVDF) film

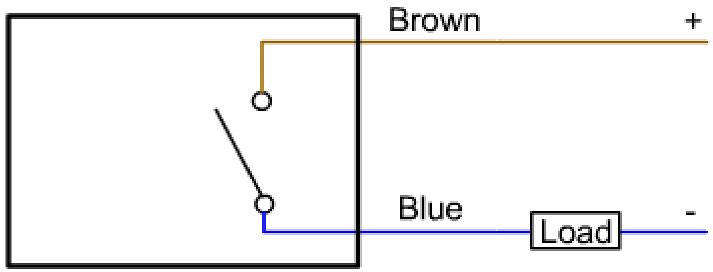
- Two layers of the film are separated by a soft film which transmits vibrations
- An ac voltage applied to the lower PVDF and this results in mechanical oscillations due to the piezoelectric effect.
- The intermediate film transmits these vibrations to the upper PVDF film.
- As a consequence of the piezoelectric effect, vibrations cause an alternating voltage to be produced across the upper film.

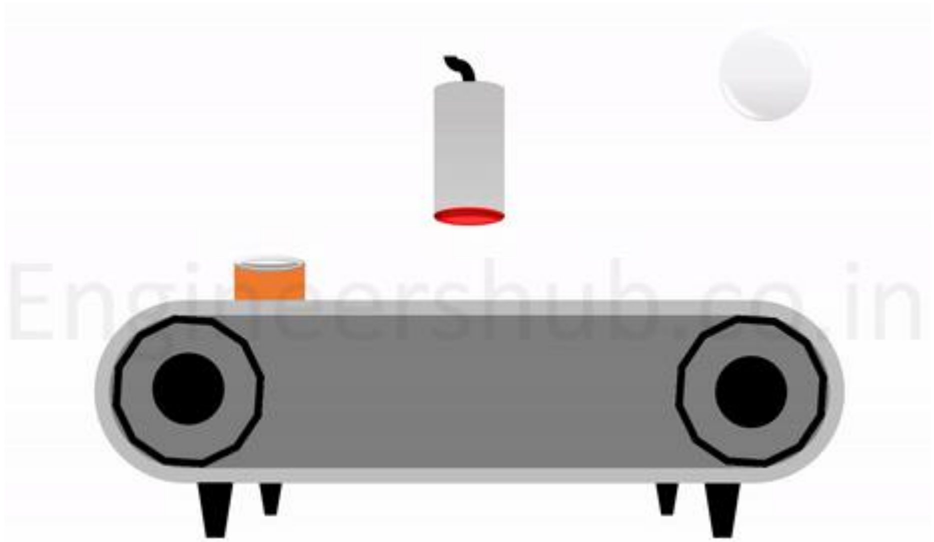
When pressure is applied to the upper PVDF film its vibrations are affected and the output alternating voltage is changed



Proximity switch or Proximity sensor

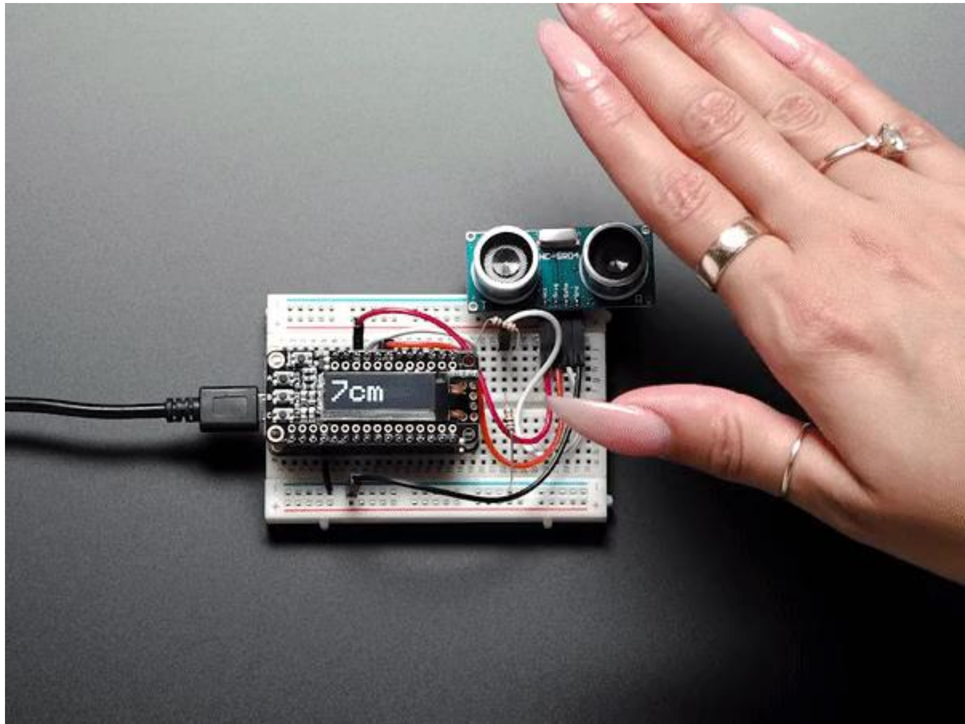
- A proximity switch, also sometimes called a proximity sensor, is a sensor that *detects the presence or absence of an object without physically contacting it*
- There are several types of proximity switches, each using a different principle for non-contact detection:
  - Inductive: Detects metal objects using a magnetic field.
  - Capacitive: Detects any material (metal, plastic, liquid) by sensing changes in capacitance.
  - Ultrasonic: Detects objects through sound wave emission and echo analysis.
  - Photoelectric: Detects objects using light beams.





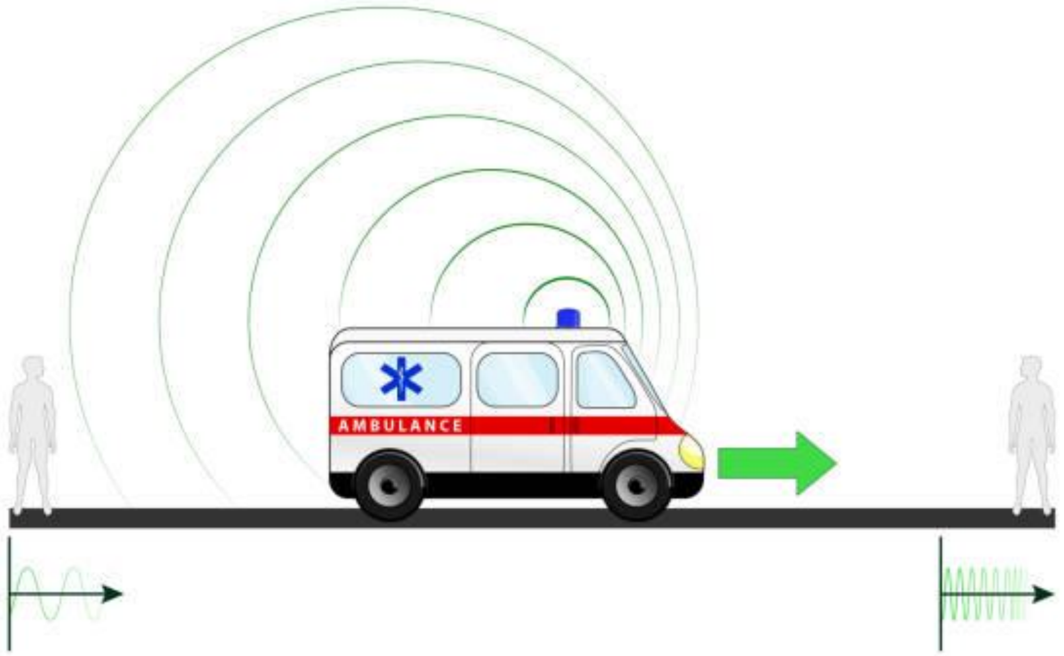
Acoustic sensor : sonar distance sensor

- A sonar distance sensor is a type of proximity sensor that uses sound waves to detect and measure the distance of an object.
- It works similarly to how bats and dolphins navigate using echolocation
- Distance = (Speed of sound * Time taken for round trip) / 2



Acoustic sensor : Doppler effect sensor

- A Doppler effect sensor is not a single specific sensor type, but rather a category of sensors that *utilize the Doppler effect to detect and measure various things*
- The Doppler effect is the principle that the frequency of a wave changes when the source of the wave and the observer are moving relative to each other.
- Imagine an ambulance approaching you with its siren. The sound gets higher-pitched (higher frequency) as it approaches and lower-pitched (lower frequency) as it moves away.



Actuation systems

Introduction

- Actuation systems are the elements of control systems which are responsible for transforming the output of a microprocessor or control system into a controlling action on a machine or device.

  Example : an electrical output from the controller which has to be transformed into a linear motion to move a load.

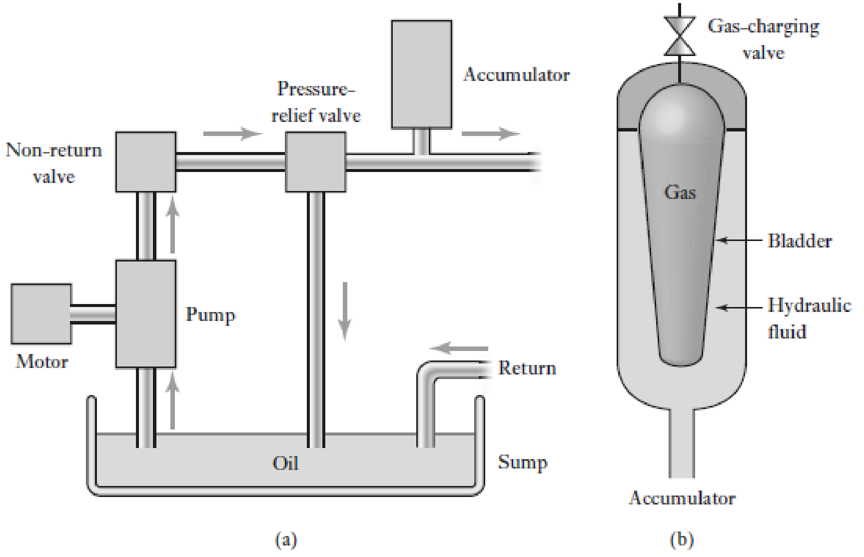
  Example : an electrical output from the controller has to be transformed into an action which controls the amount of liquid passing along a pipe.

- ❖ Pneumatic and hydraulic actuation systems
- ❖ mechanical actuator systems
- ❖ electrical actuation systems

Hydraulic actuation systems

- Hydraulic system uses pressurized oil is provided by a pump driven by an electric motor

Figure 7.1 (a) Hydraulic power supply, (b) accumulator.



- The pump, pumps oil from a sump through a non-return valve and an accumulator to the system, from which it returns to the sump
- A *pressure-relief valve releases the pressure if it rises above a safe level*, the *non-return valve is to prevent the oil being back driven to the pump* and the *accumulator is to smooth out any short-term fluctuations in the output oil pressure*.
- Pressurized gas supplied to the bladder and the chamber containing the hydraulic fluid
- If the oil pressure rises then the bladder contracts, increases the volume the oil can occupy and so reduces the pressure.
- If the oil pressure falls, the bladder expands to reduce the volume occupied by the oil and so increases its pressure.

HYDRAULIC PUMPS

- Pumps are used to supply the high pressure that the mechatronic system requires.

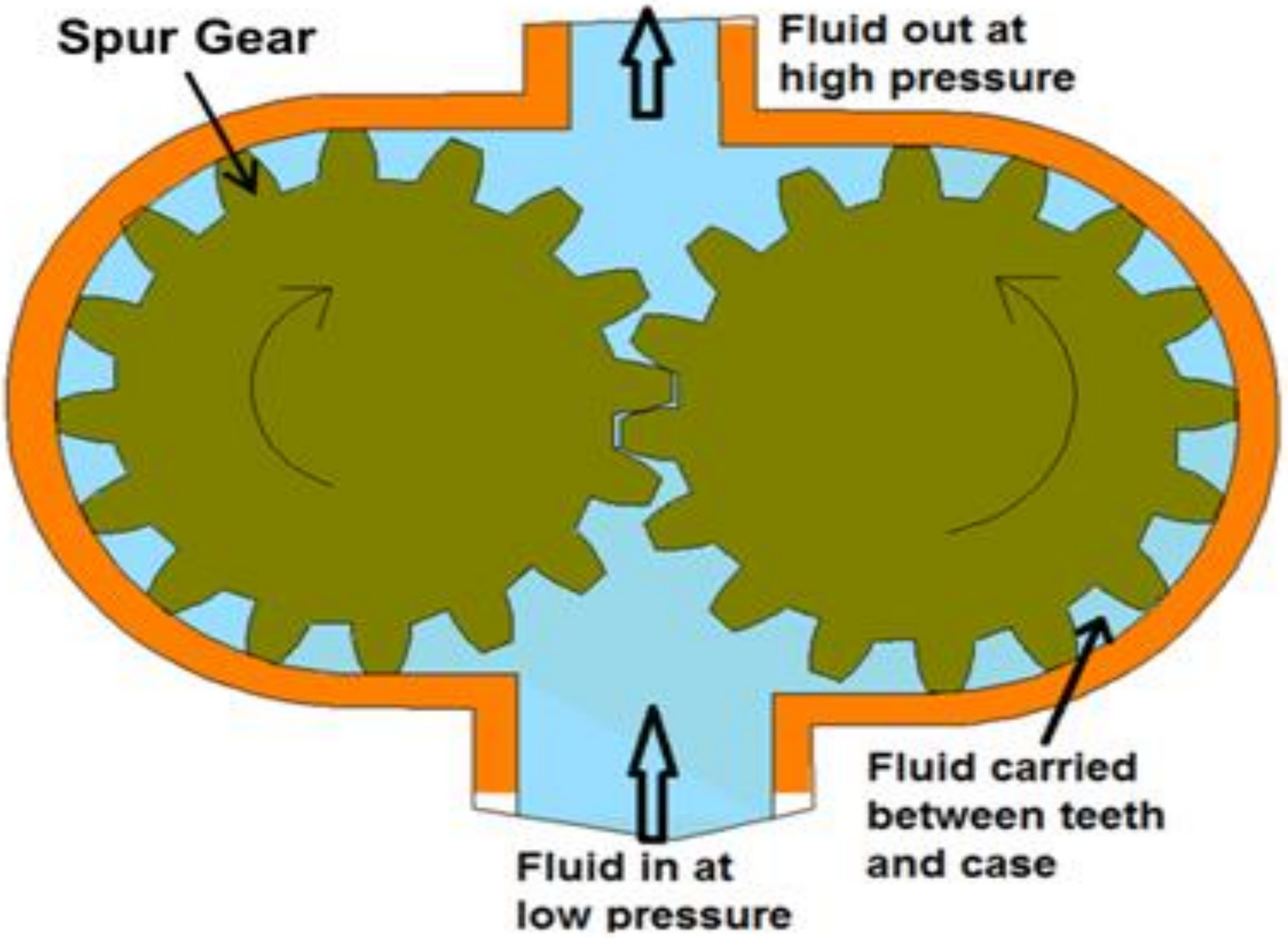
Three types are most commonly used:

  - Gear pump;
  - Vane pump; and
  - Piston pump.

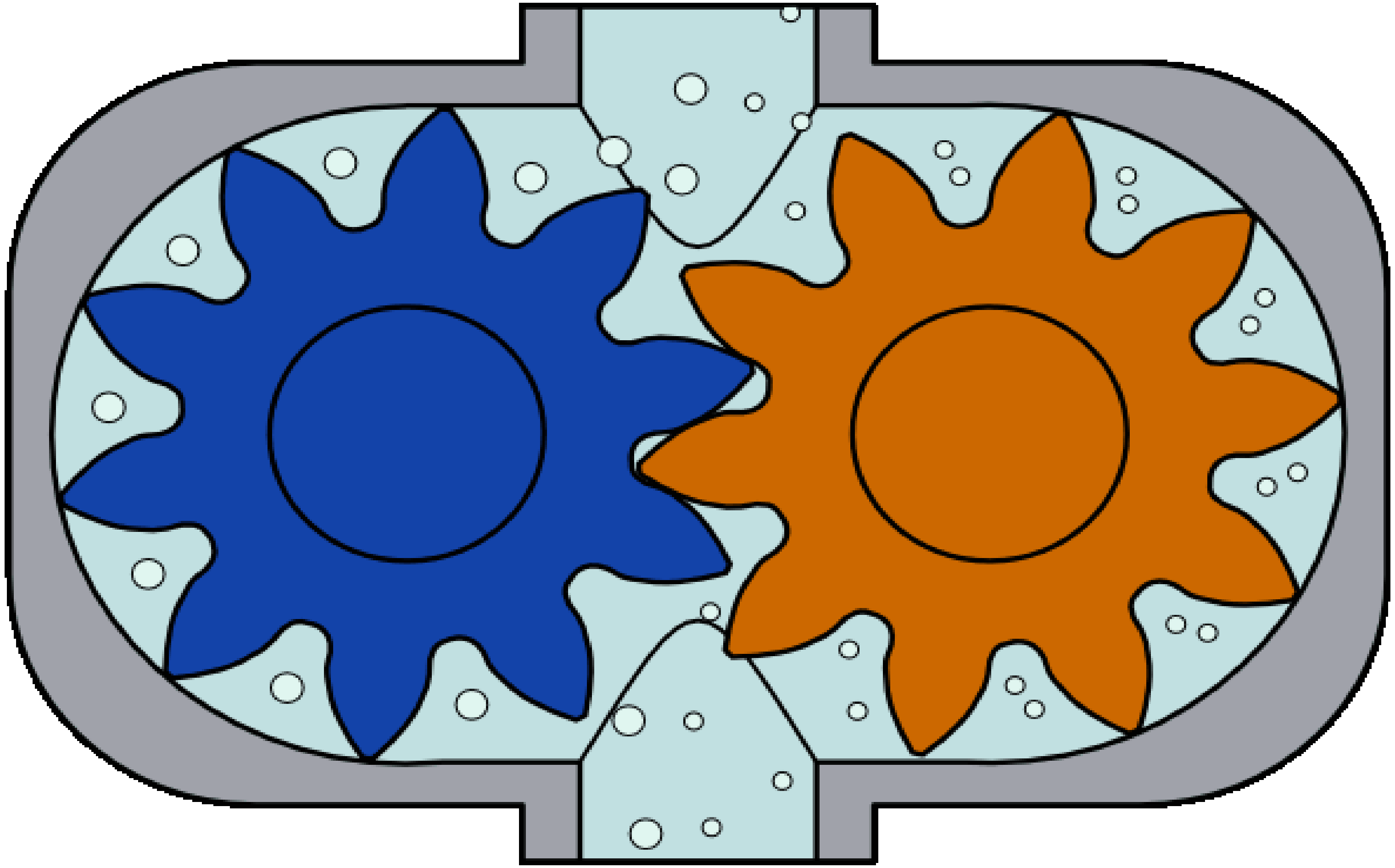
Gear Pump:

- Two close-meshing gear wheels which rotate in opposite directions.
- One of the gears is coupled with a prime mover and is called as driving gear and another is called as driven gear.
- When the gears rotate, volume of the chamber expands leading to pressure drop below atmospheric value.
- Therefore the vacuum is created and the fluid is pushed into the void due to atmospheric pressure.
- The fluid is trapped between housing and rotating teeth of the gears.
- The leakage occurs between the teeth and the casing and between the interlocking teeth limits the efficiency

Gear Pump:


Spur Gear
Fluid out at
high pressure
Fluid carried
between teeth
and case
Fluid in at
low pressure

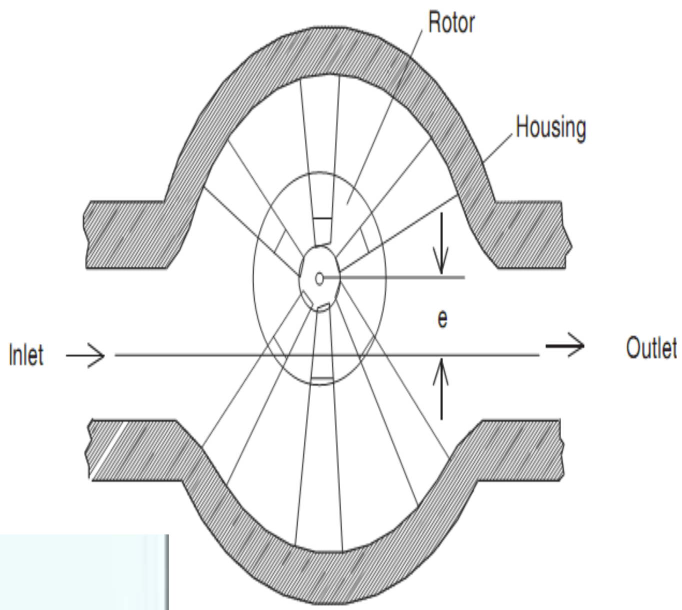
Gear Pump:

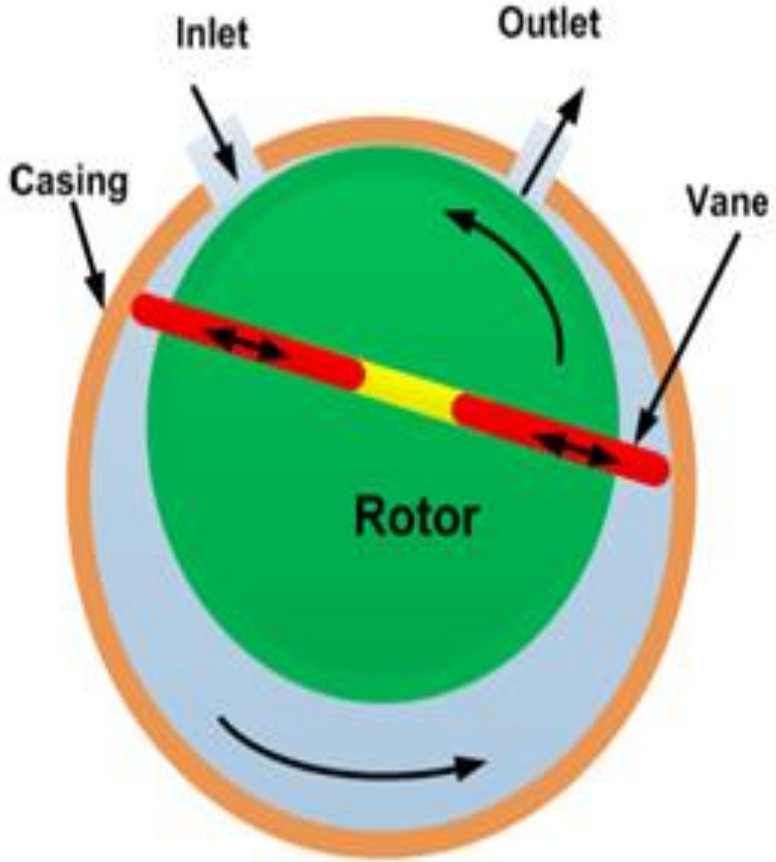


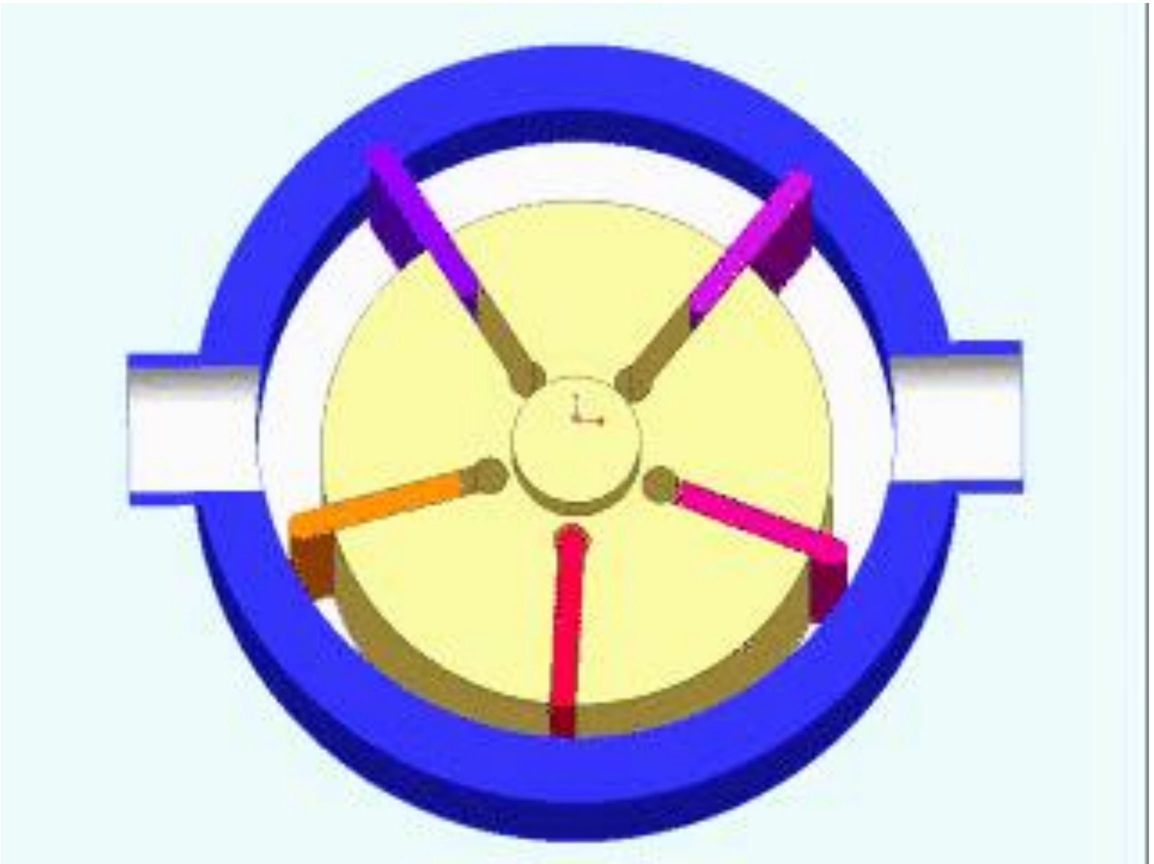
Vane Pump:

- Vane pumps generate a pumping action by tracking of vanes along the casing wall.
- The vane pump has *spring-loaded sliding vanes slotted in a driven rotor*
- The vane pumps generally consist of a rotor, vanes, ring and a port plate with inlet and outlet ports.
- The rotor in a vane pump is connected to the prime mover through a shaft. The vanes are located on the slotted rotor.
- *As the rotor rotates, the vanes follow the contours of the casing. This results in fluid becoming trapped between successive vanes and the casing and transported round from the inlet port to outlet port.*
- The leakage is less than with the gear pump.

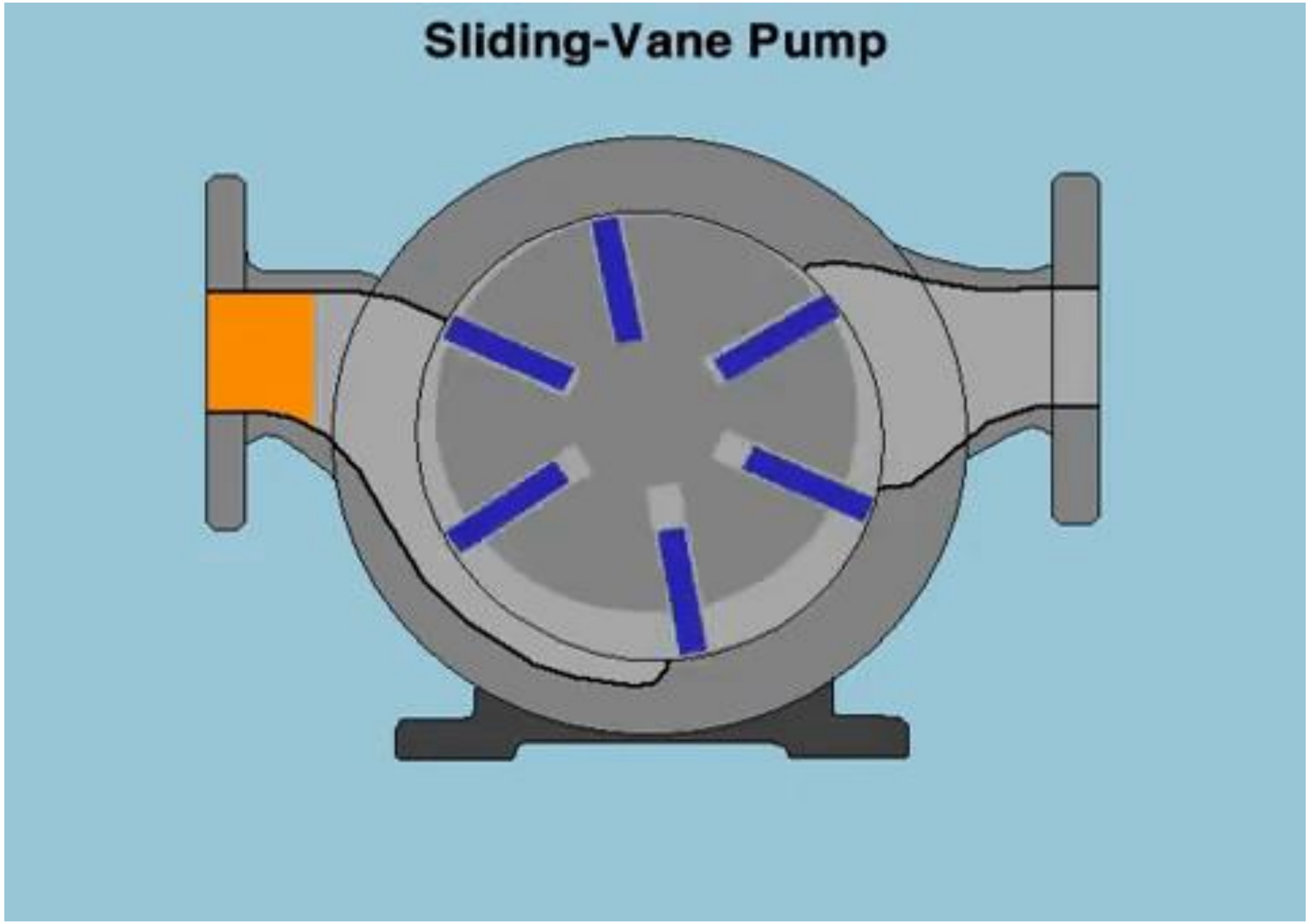
Vane Pump:


Rotor
Housing
e
Inlet
Outlet


Inlet
Outlet
Casing
Vane
Rotor

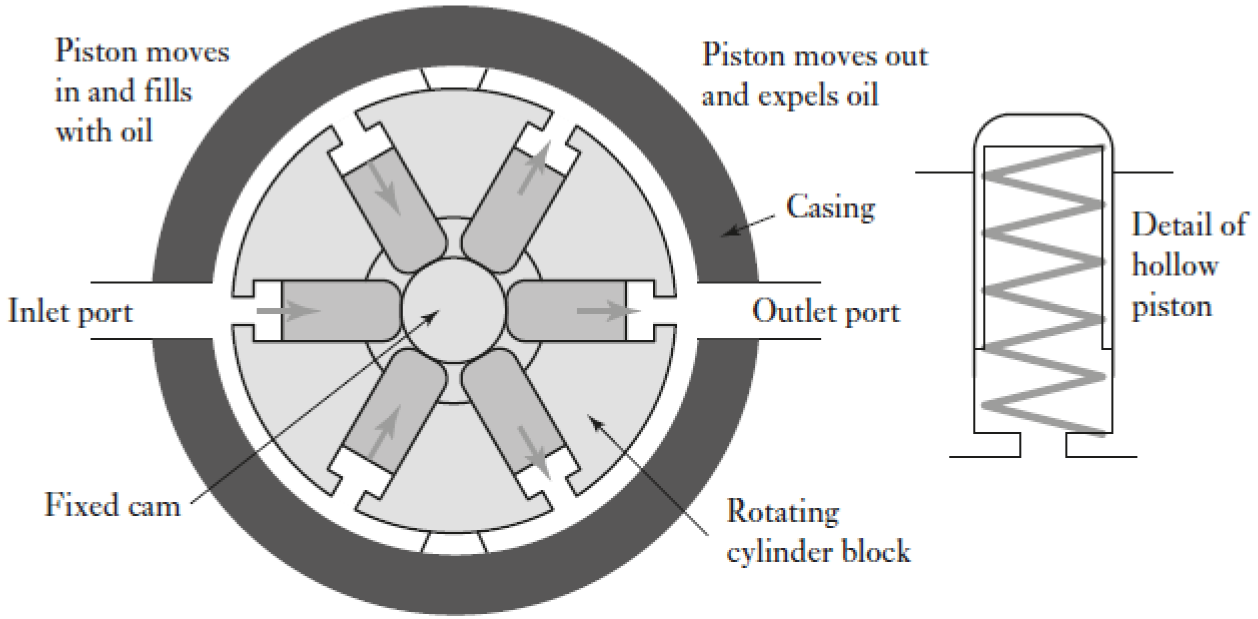


Vane Pump:



Piston Pump:

Radial piston pump

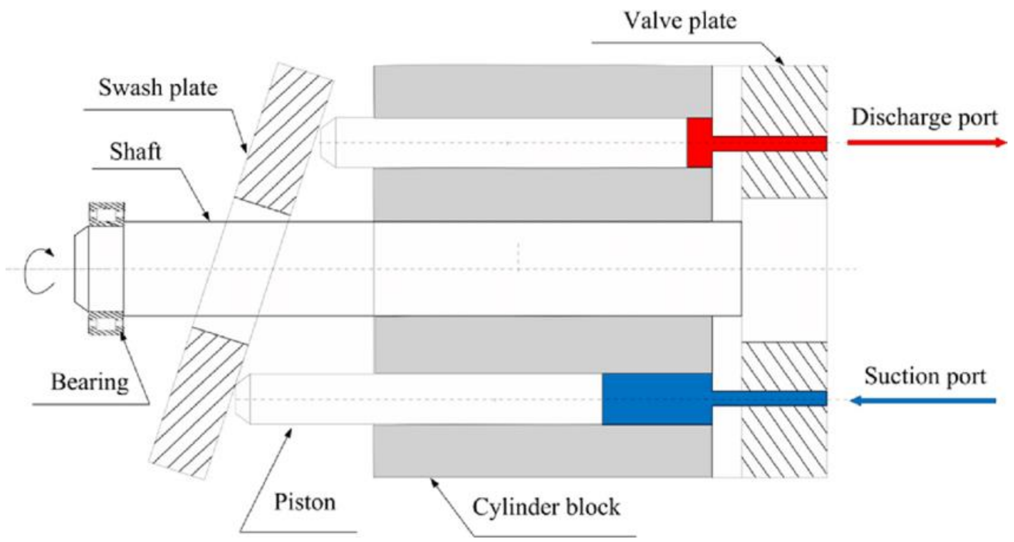


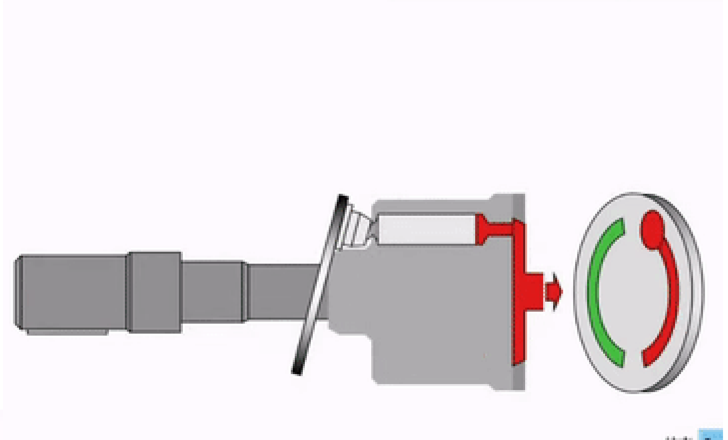
- a cylinder block rotates round the stationary cam
- Hollow pistons, with spring return, to move in and out
- The fluid is drawn through the inlet port and transported round and ejection through the discharge port

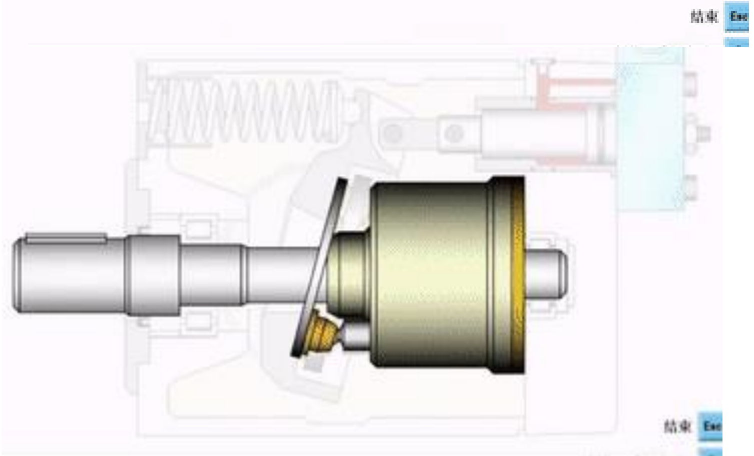
Piston Pump

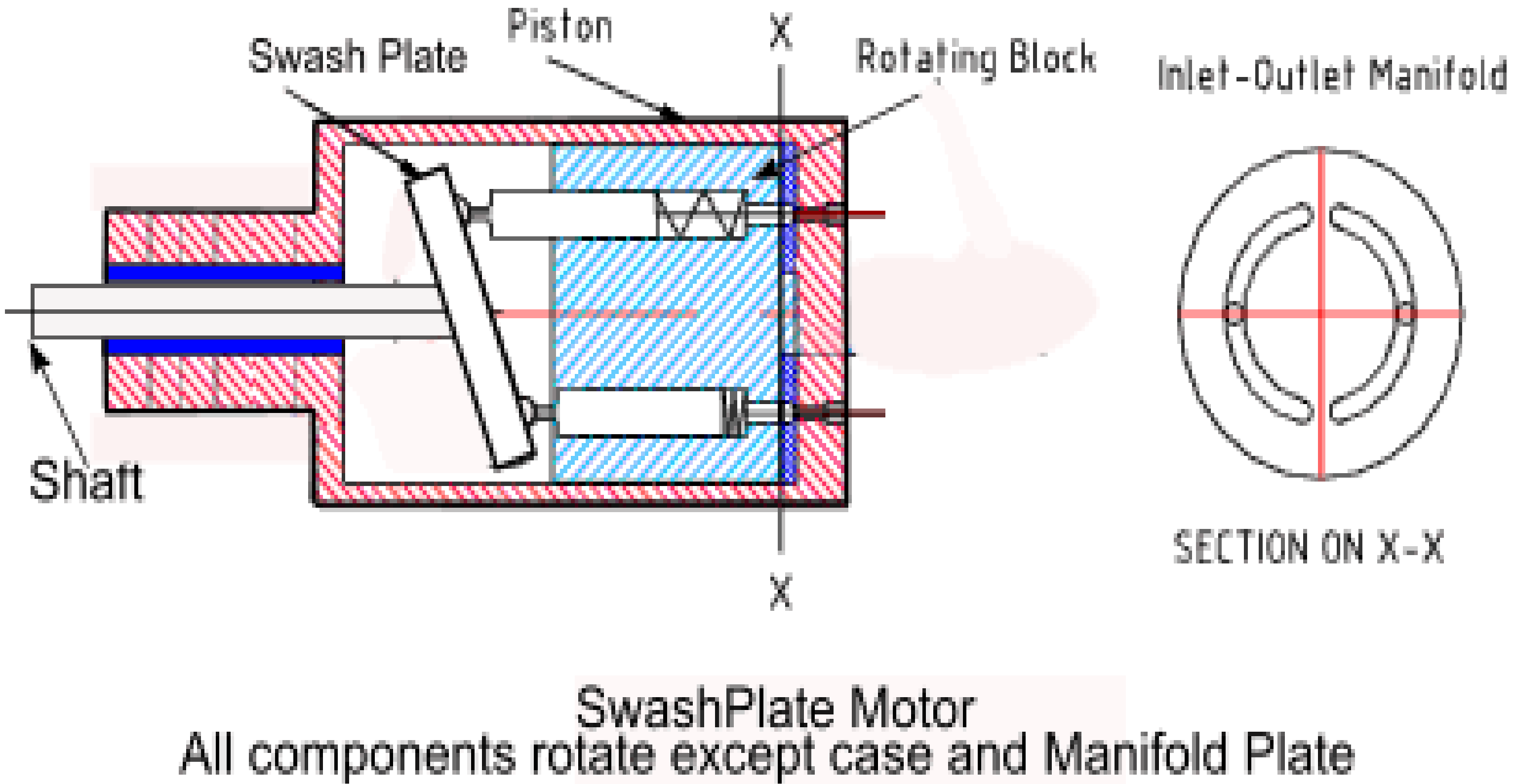
<u>Axial piston pump</u>

- Pistons which move axially rather than radially





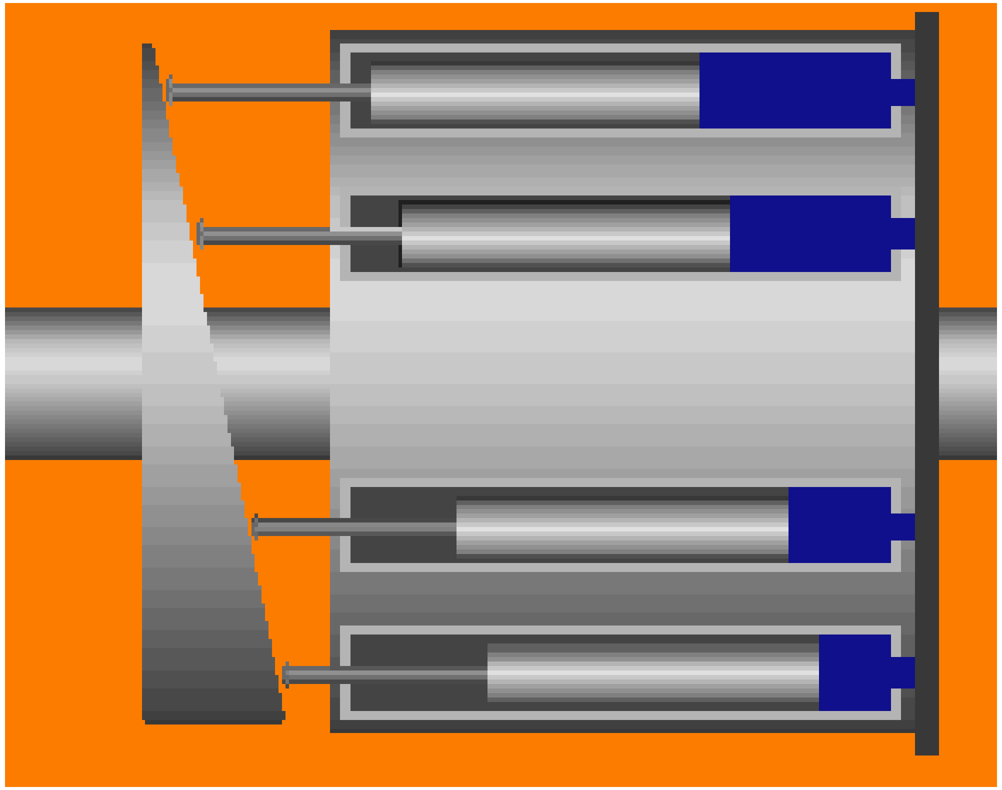


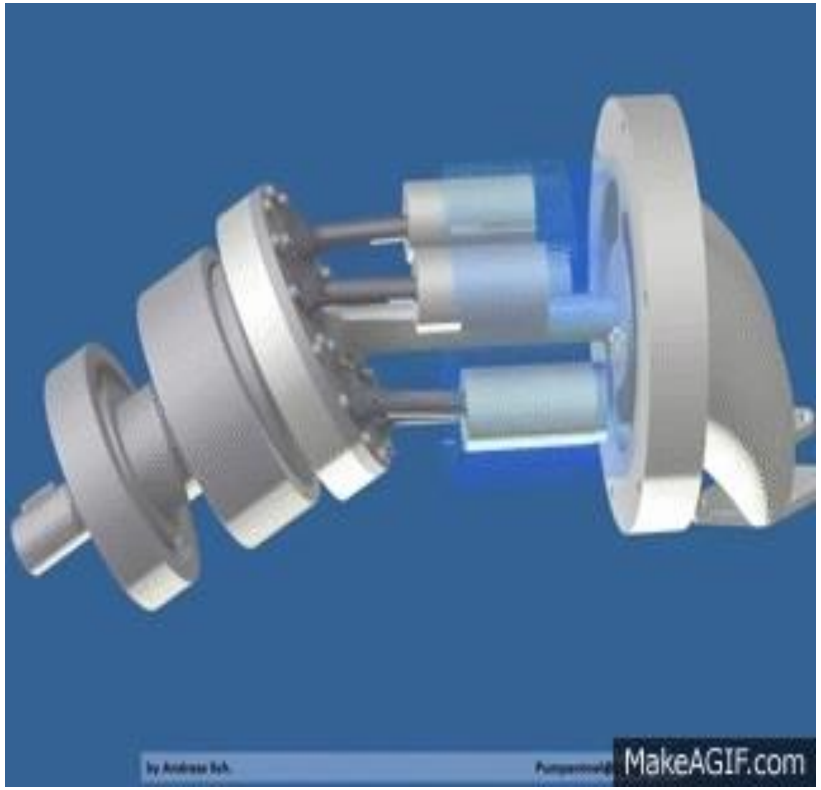


Axial piston pump

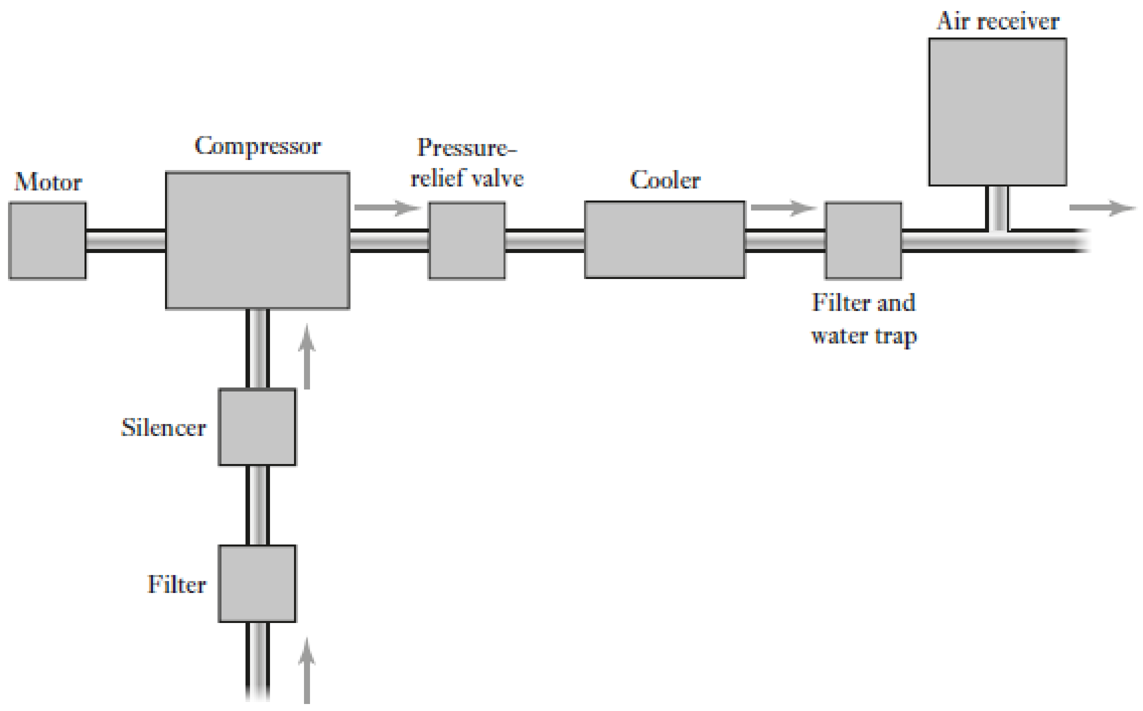
- Axial piston pumps are positive displacement pumps which converts rotary motion of the input shaft into an axial reciprocating motion of the pistons.
- These pumps have a number of pistons (usually an odd number) in a circular array within a housing which is commonly referred to as a cylinder block, rotor or barrel.
- Pisons are arranged axially in a rotating cylinder block and made to move by contact with the swash plate.
- This plate is at an angle to the drive shaft
- The shaft rotates they move the pistons so that air is sucked in when a piston is opposite the inlet port and expelled when it is opposite the discharge port.
- Piston pumps have a high efficiency and can be used at higher hydraulic pressures than gear or vane pumps.

Axial Piston Pump:





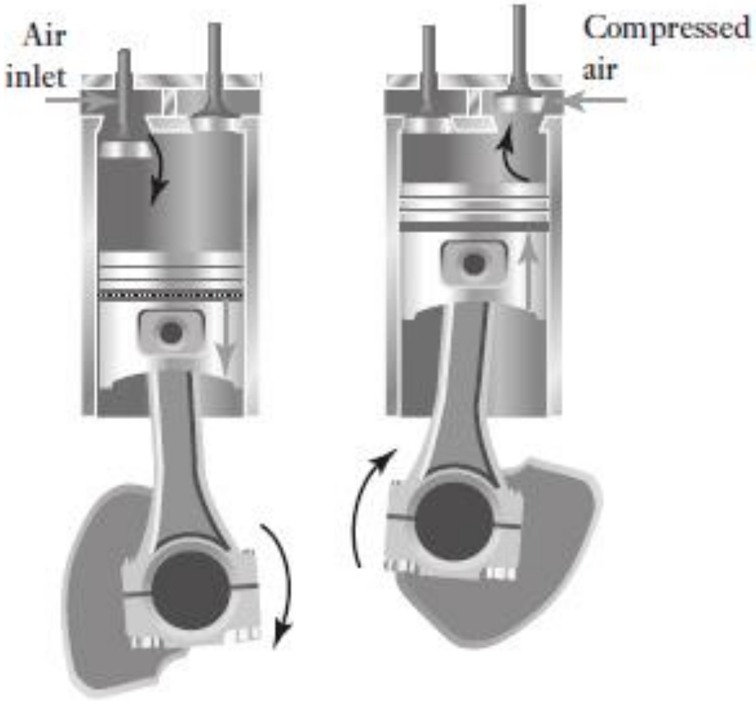
Pneumatic actuation systems



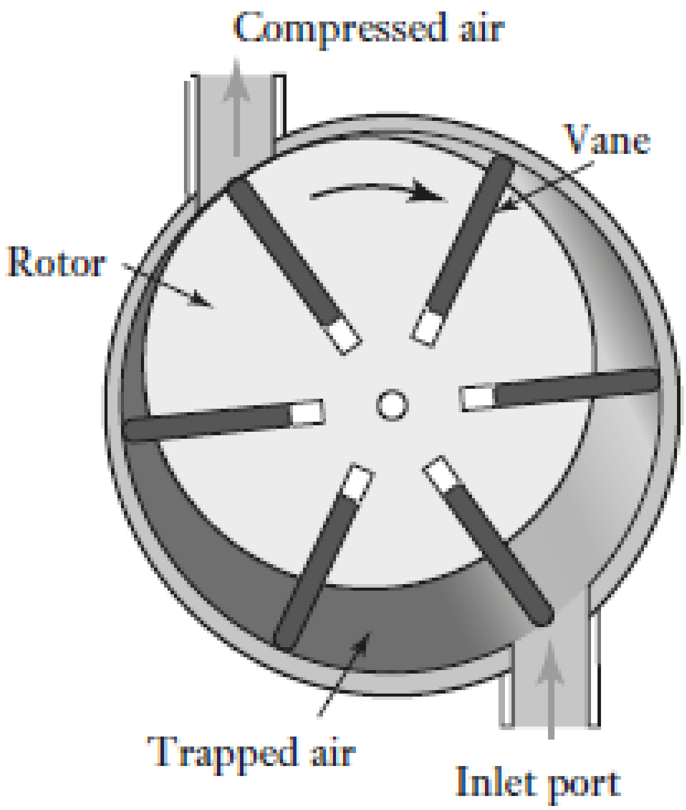
- Working with a pneumatic power supply
- An electric motor drives an air compressor
- The air inlet to the compressor is likely to be filtered and via a silencer to reduce the noise level.
- A pressure-relief valve provides protection against the pressure in the system rising above a safe level.
- The air compressor increases the temperature of the air, there is likely to be a cooling system
- To remove contamination and water from the air a filter with a water trap.
- An air receiver smooth-out any short-term pressure fluctuations.

Single-acting single-stage reciprocating compressor

- The air intake stroke, the descending piston causes air to be sucked into the chamber through the spring-loaded inlet valve
- The trapped air forces the inlet valve to close and so air becomes compressed.
- When the air pressure has risen sufficiently, the spring-loaded outlet valve opens and the trapped air flows into the compressed-air system.
- Pressure up to140 bar, air flow deliveries range from 0.02 m^3/min to 600 m^3/min

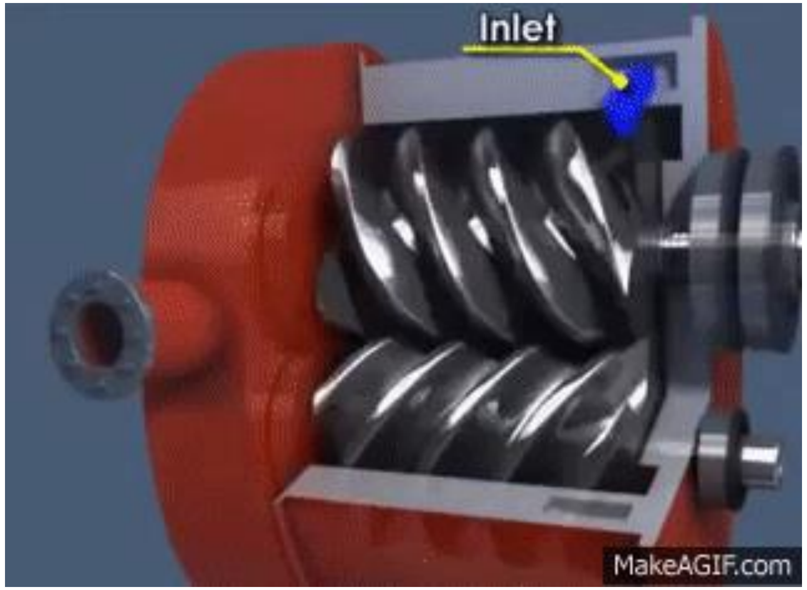


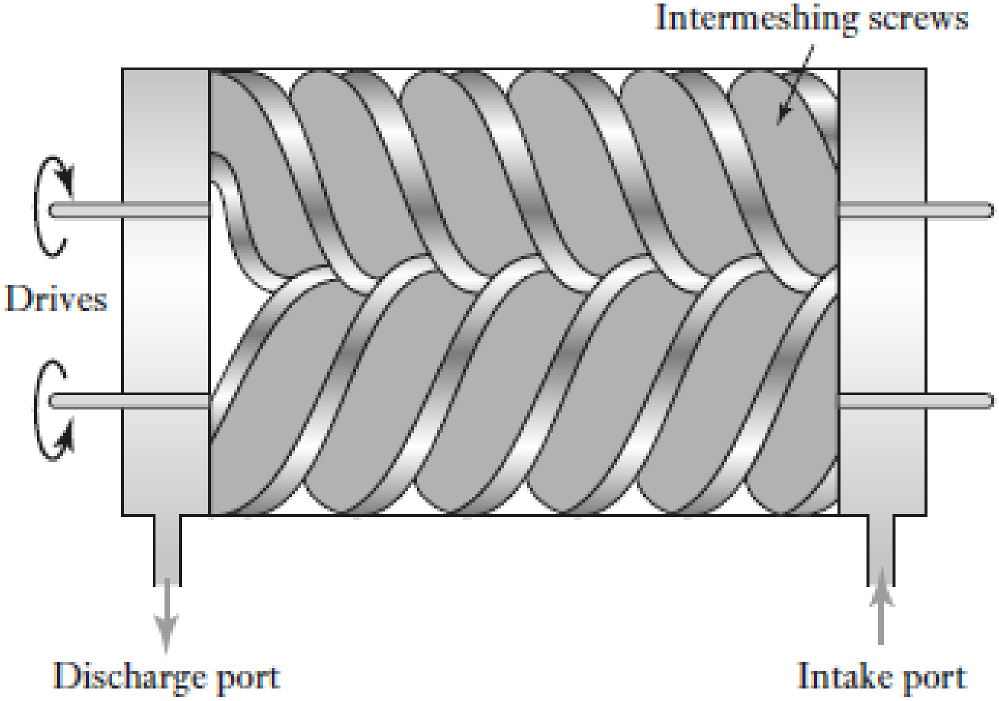
Rotary vane compressor



- Rotor mounted eccentrically in a cylindrical chamber
- The rotor has blades, the vanes, which are free to slide in radial slots with rotation causing the vanes to be driven outwards against the walls of the cylinder
- As the rotor rotates, air is trapped in pockets formed by the vanes and as the rotor rotates so the pockets become smaller and the air is compressed.
- Compressed air are thus discharged from the discharge port.
- Pressures up to about 800 kPa with flow rates of the order of 0.3 m^3/min to 30 m^3/min

Rotary screw compressor

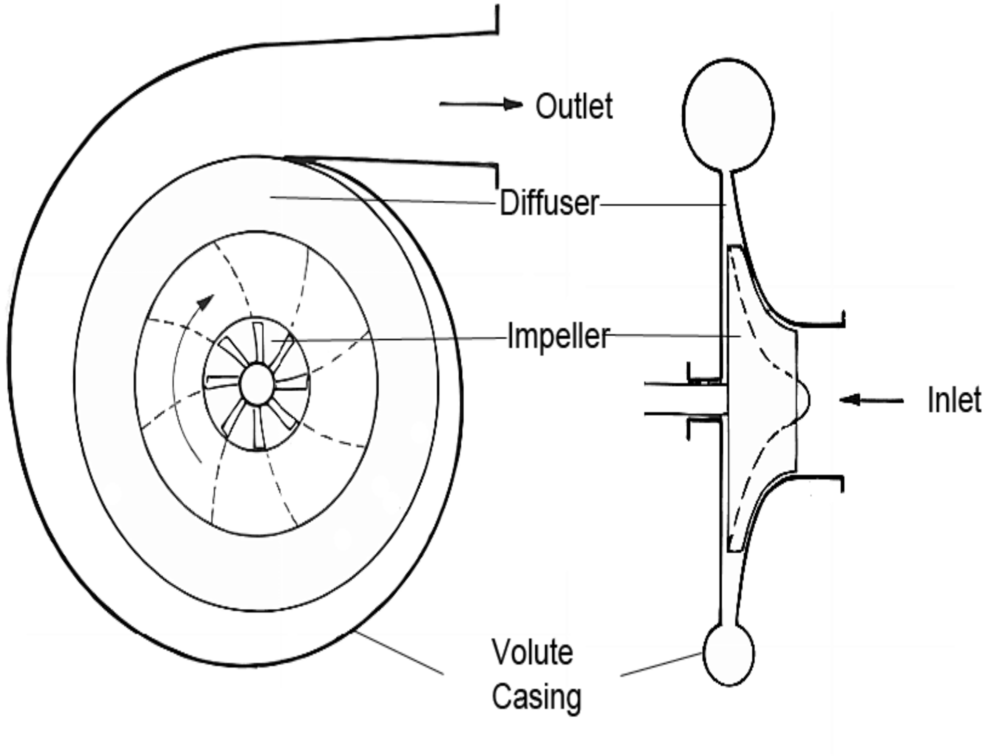




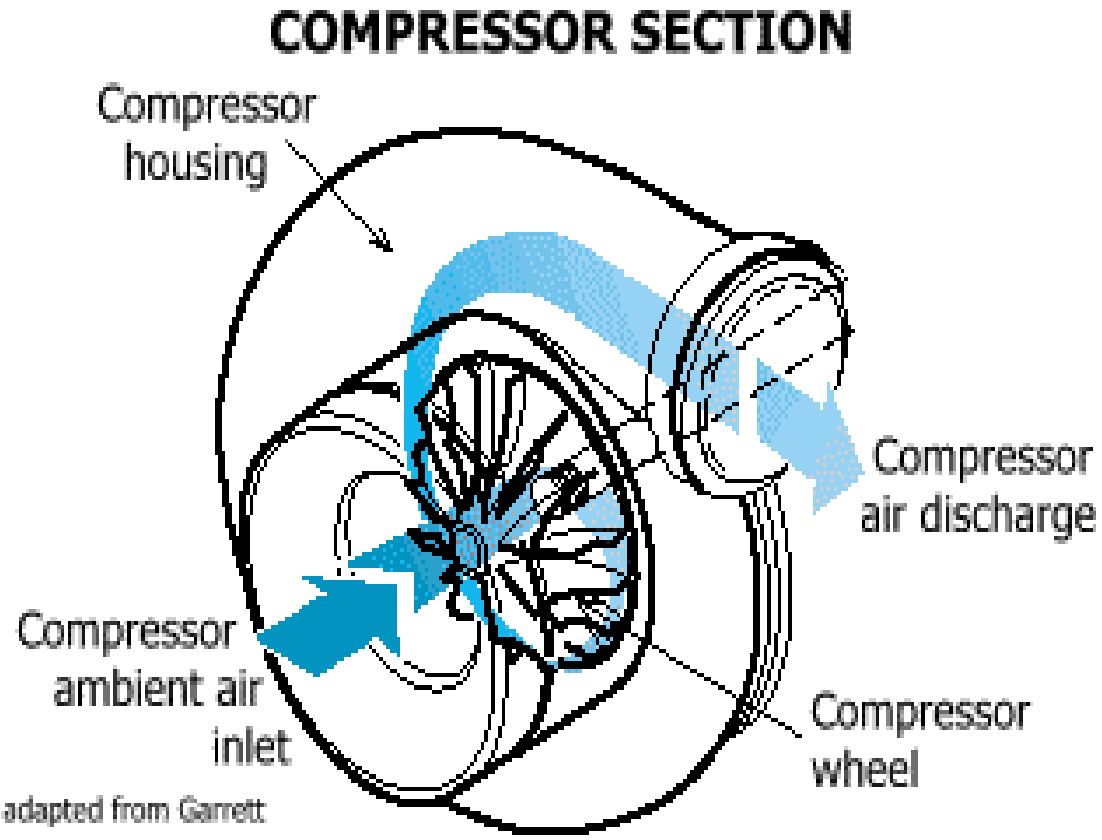
- This has two intermeshing rotary screws which rotate in opposite directions.
- As the screws rotate, air is drawn into the casing through the inlet port and into the space between the screws
- This trapped air is moved along the length of the screws and compressed as the space becomes progressively smaller
- Emerging from the discharge port.
- Pressures up to about 1000 kPa with flow rates of between 1.4 m^3/min and 60 m^3/min

Centrifugal compressors

- In centrifugal compressors, air enters the eye or centre of the impeller
- Centrifugal effect throws the air into the volute where it goes to the diffuser

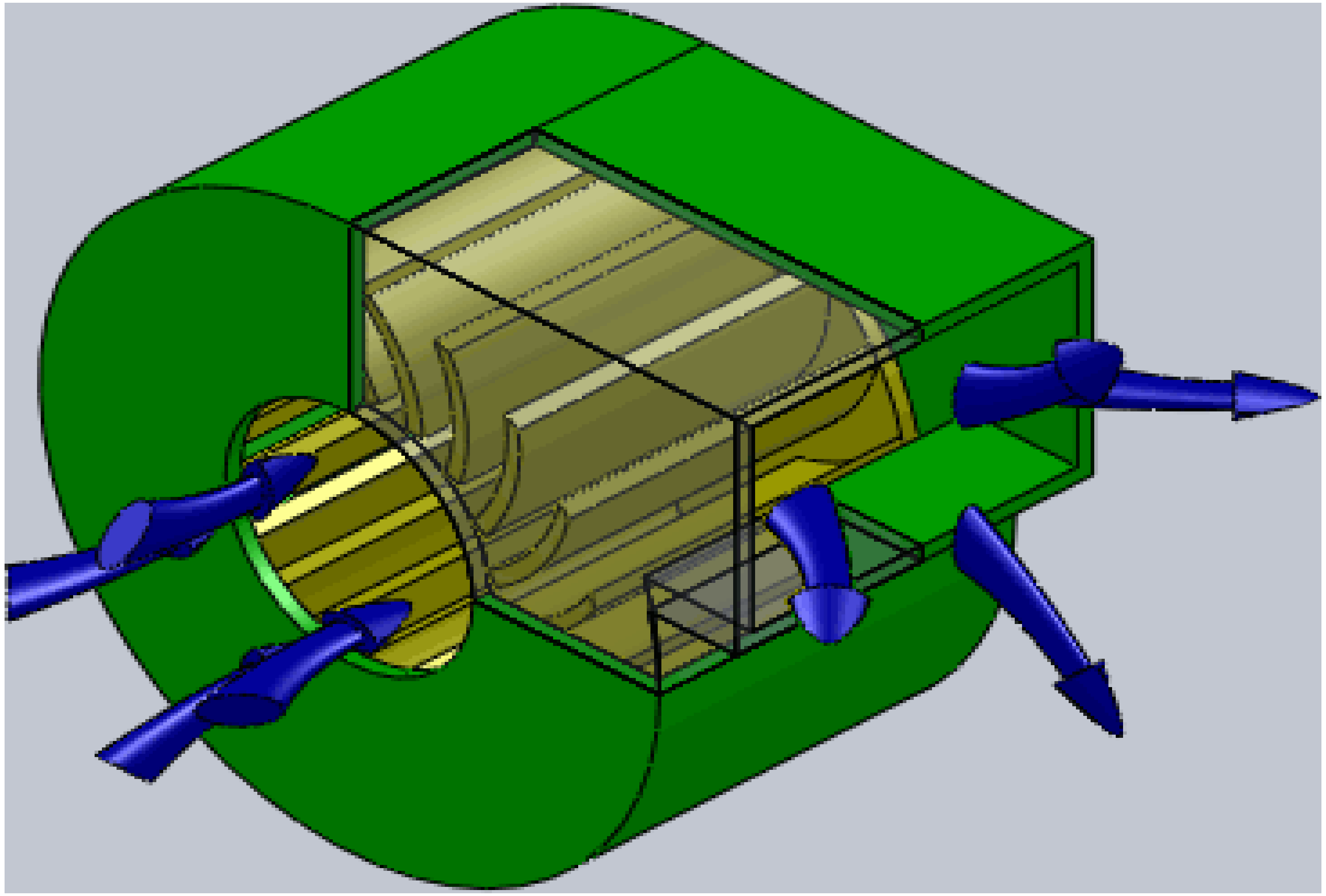


Centrifugal compressor schematic diagram



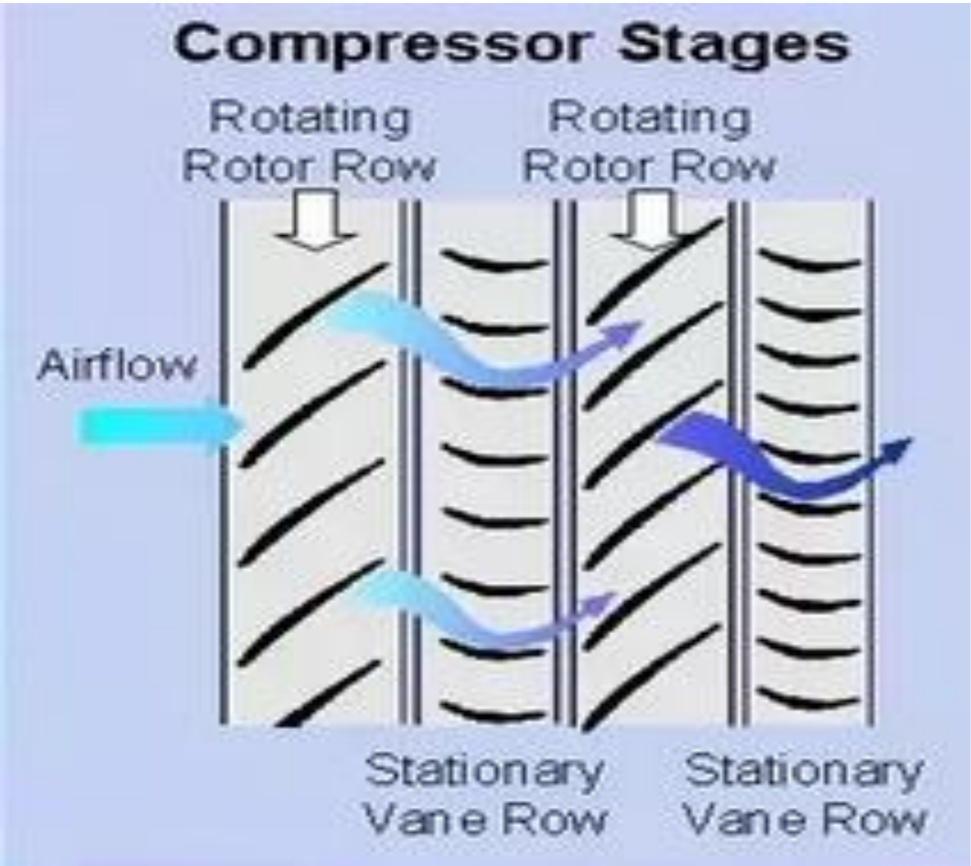
adapted from Garrett

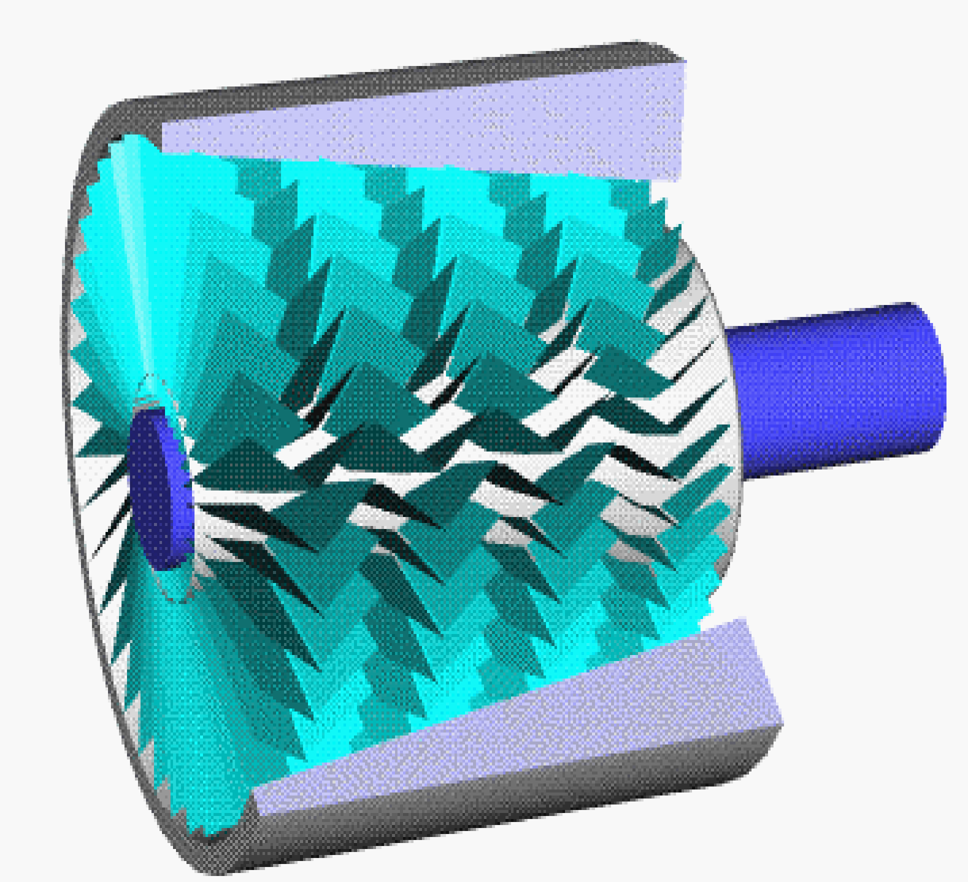
Centrifugal compressors



Axial compressor

- In axial compressors, the annular inlet area is much greater than that of the eye of the centrifugal type. Hence it delivers more flow.
- The blades are attached to the tapered rotor.





Linear Hydraulic Actuators - Cylinders

- Hydraulic linear actuators, as their name implies, provide motion in a straight line.
- They are usually referred to as cylinders, rams and jacks.
- ***The function of hydraulic cylinder is to convert hydraulic power into linear mechanical force or motion.***
- Hydraulic cylinders extend and retract a piston rod to provide a push or pull force to drive the external load along a straight-line path.

Types of Linear Hydraulic Actuators:

Hydraulic cylinders are of the following types:

- Single-acting cylinders
- Double-acting cylinders
- Telescopic cylinders.
- Tandem cylinders.

Single Acting Cylinders:

- A single-acting cylinder is simplest in design.
- *These cylinders use hydraulic pressure to extend the piston rod in one direction and rely on an external force, such as a spring, to retract it.*

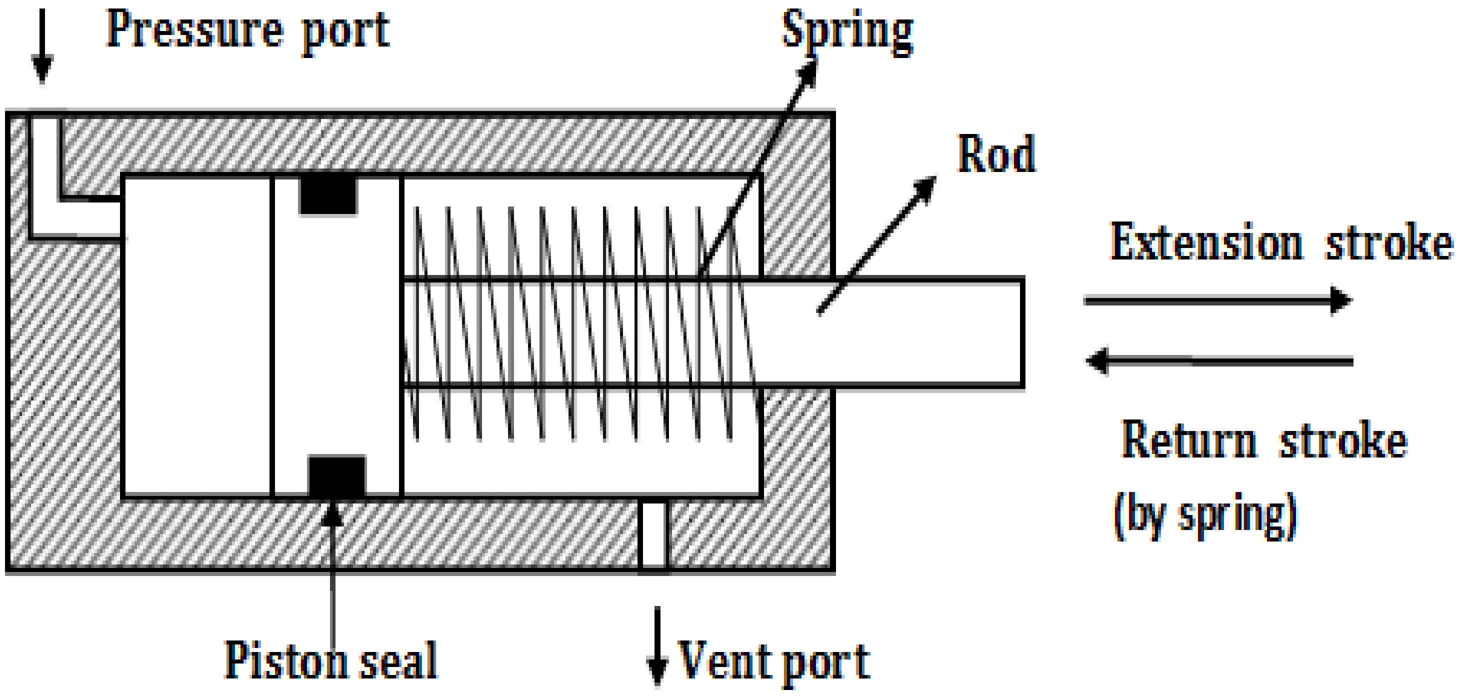


Figure 1.1 Single-acting cylinders

According to the type of return, single-acting cylinders are classified as follows:

- Gravity-return single-acting cylinder.
- Spring-return single-acting cylinder.

Gravity-Return Single-Acting Cylinder

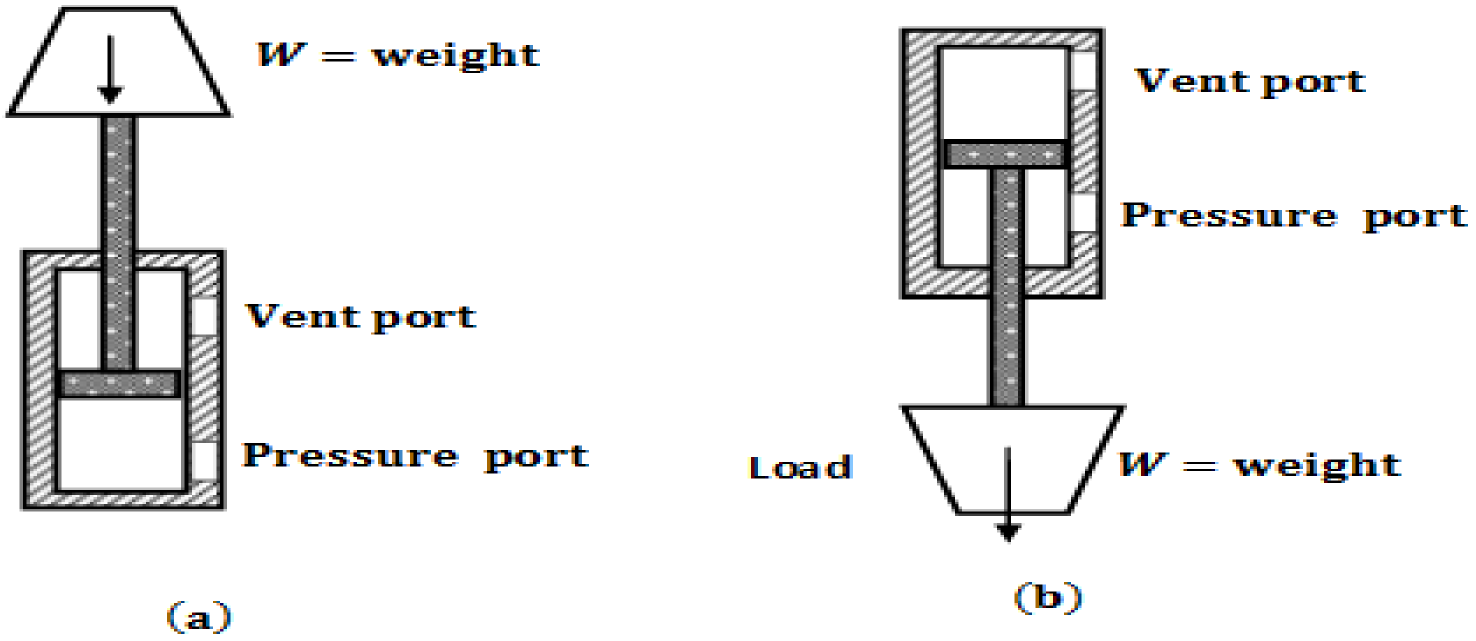


Figure 1.2 Gravity-return single-acting cylinder: (a) Push type; (b) pull type

• Spring Return Single Acting Cylinder:

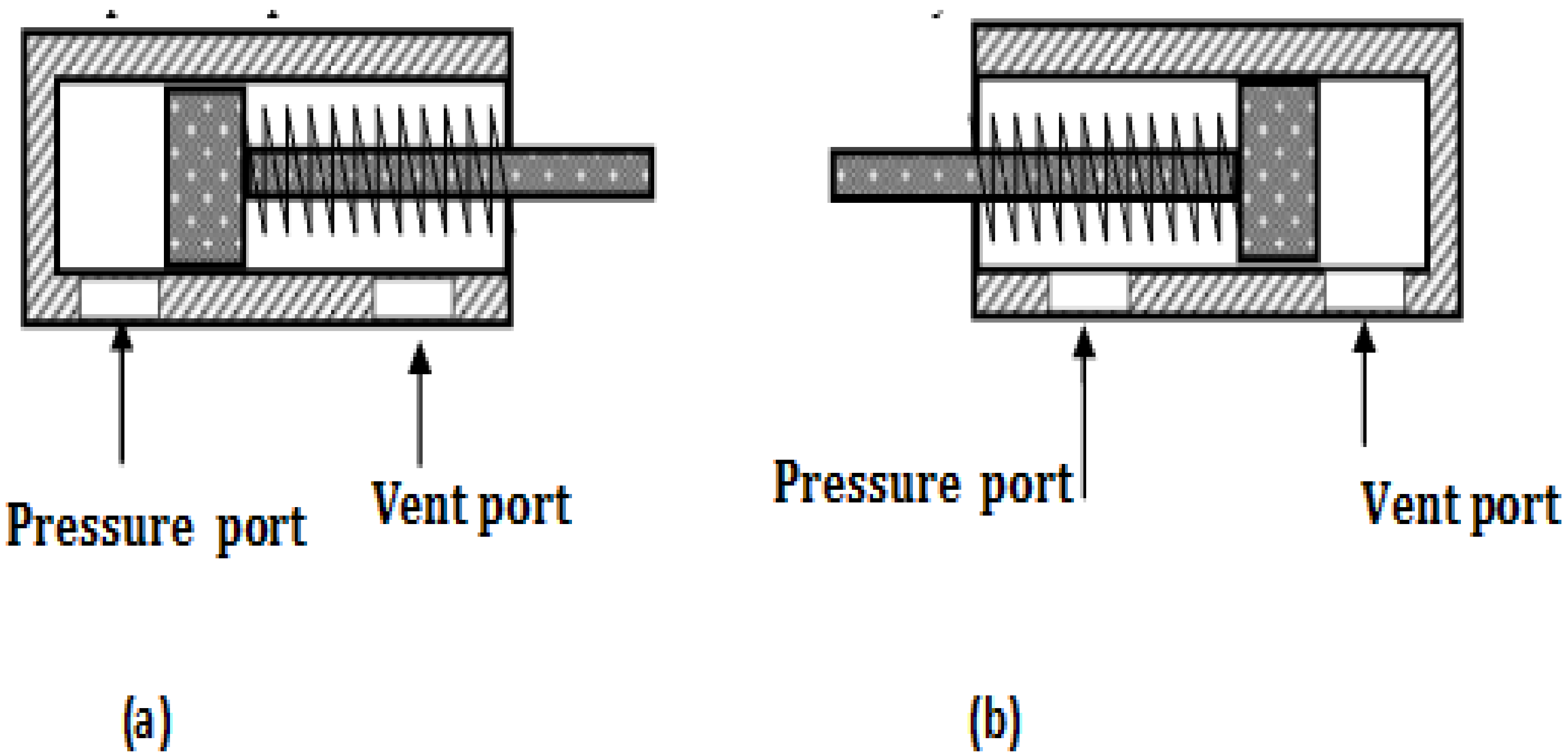


Figure 1.3 (a) Push- and (b) pull-type single-acting cylinders

Double-Acting Cylinder:

There are two types of double-acting cylinders:

- **Double-acting cylinder with a piston rod on one side:**
- Double-acting cylinder with a piston rod on both sides.

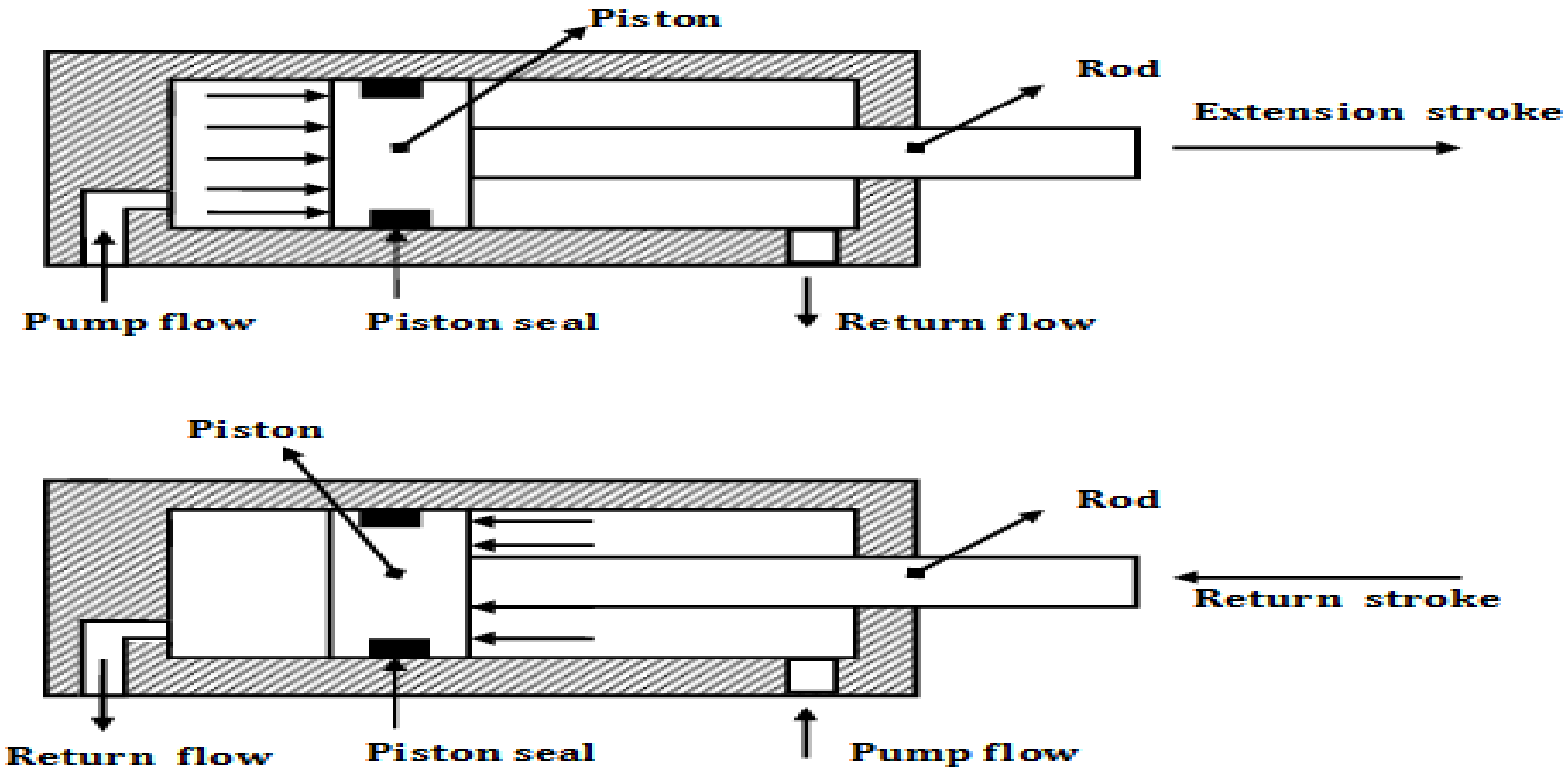


Figure 1.4 Double-acting cylinder with a piston rod on one side

Double-acting cylinder with a piston rod on both sides:

- This cylinder can be used in an application *where work can be done by both ends of the cylinder*, thereby making the cylinder more productive.

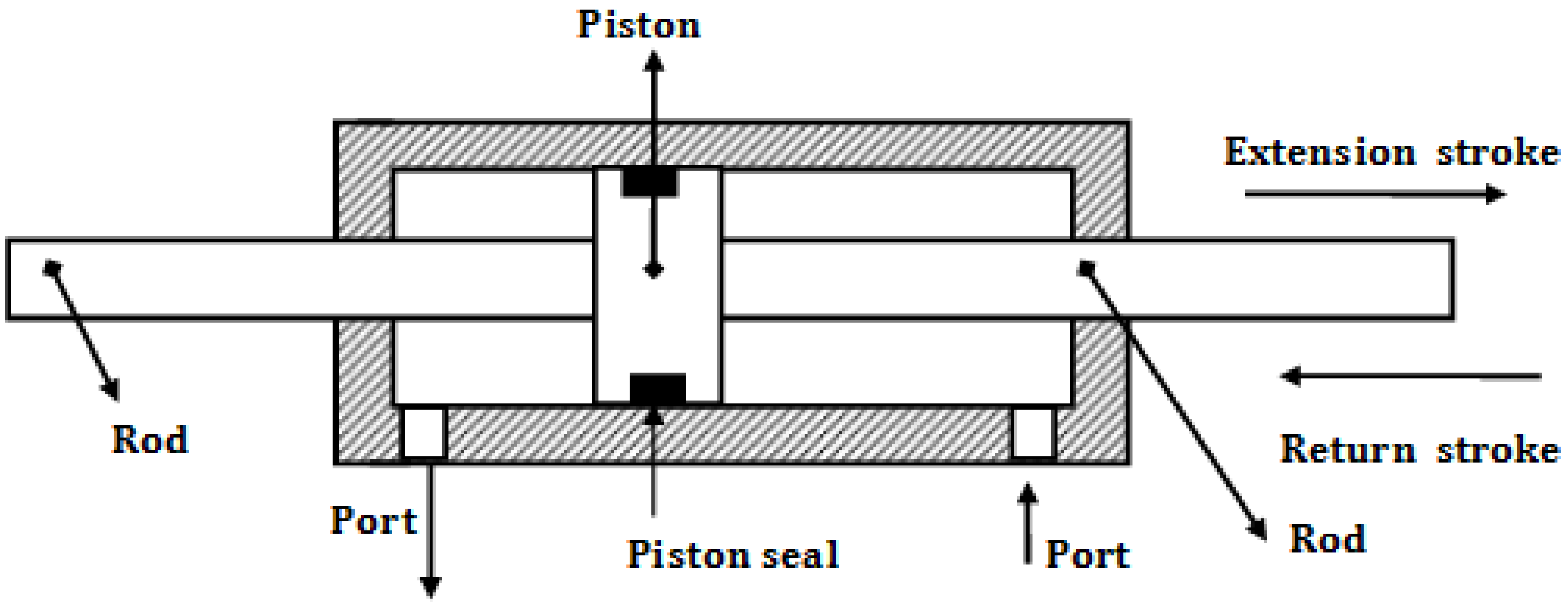


Figure 1.5Double-acting cylinder with a piston rod on one side

Telescopic Cylinder:

- A telescopic cylinder is used when a long stroke length and a short retracted length are required.
- *These cylinders have multiple stages that extend one out of the other, providing a longer stroke length than a single-stage cylinder.*
- The telescopic cylinder extends in stages, each stage consisting of a sleeve that fits inside the previous stage.
- They generally consist of a nest of tubes and operate on the displacement principle.
- The tubes are supported by bearing rings, the innermost (rear) set of which have grooves or channels to allow fluid flow.
- The front bearing assembly on each section includes seals and wiper rings. Stop rings limit the movement of each section, thus preventing separation.

Telescopic Cylinder:

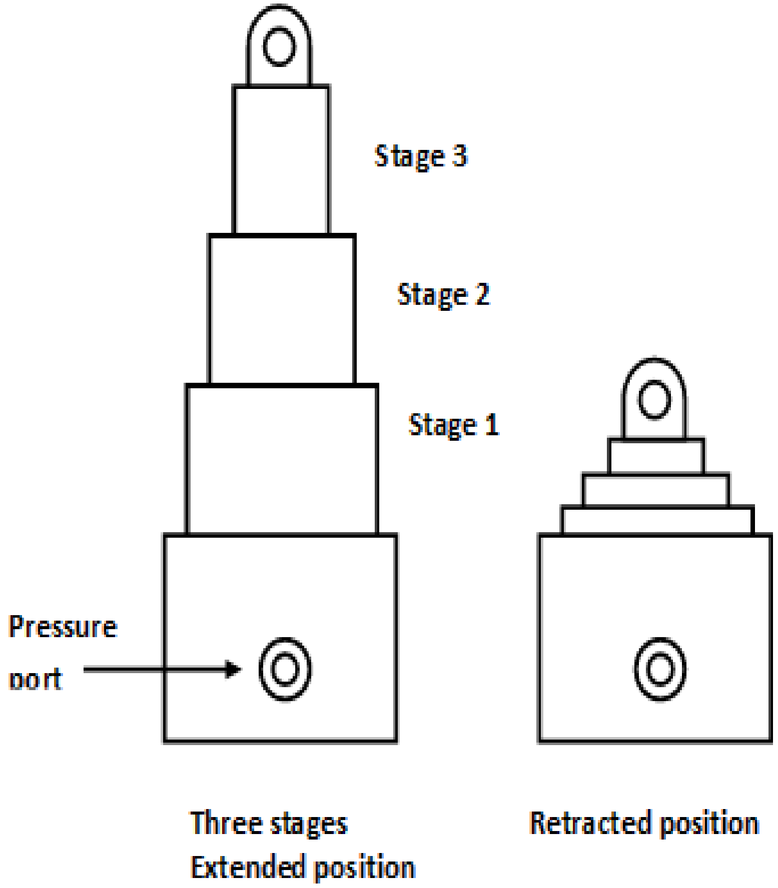
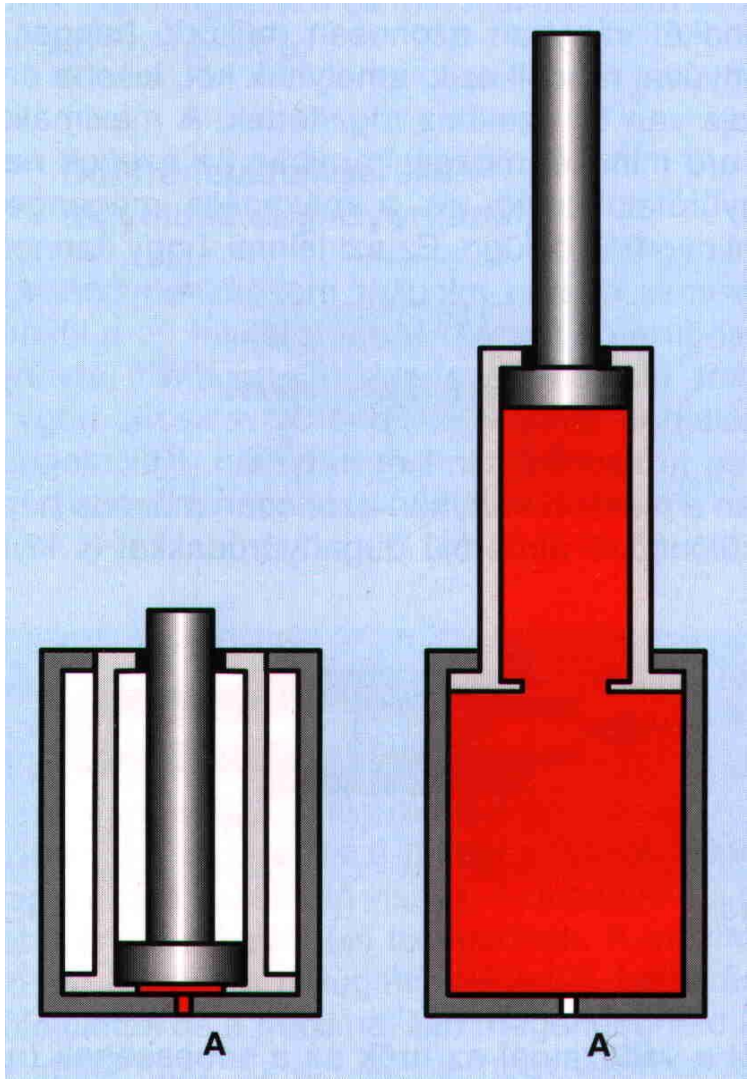


Figure 1.6 Telescopic cylinder



Tandem Cylinder:

- A tandem cylinder is used in applications where a large amount of force is required from a small-diameter cylinder.
- ***These cylinders consist of two or more single-acting cylinders plumbed together to provide increased force or stroke length.***
- They are a good option for applications where a single cylinder cannot provide the required force or stroke.
- Pressure is applied to both pistons, resulting in increased force because of the larger area.
- The drawback is that these cylinders must be longer than a standard cylinder to achieve an equal speed because flow must go to both pistons.

Tandem Cylinder:

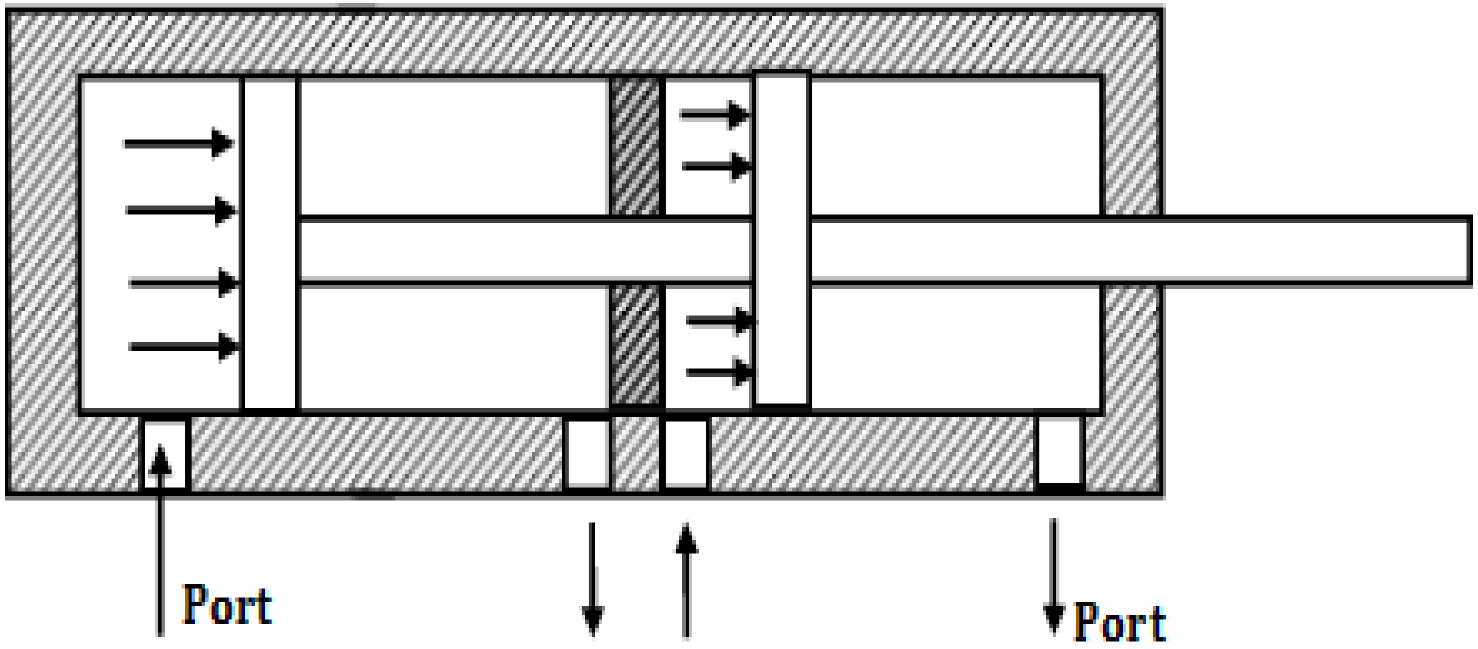


Figure 1.7Tandem cylinder